

Dalek round-body design and assembly

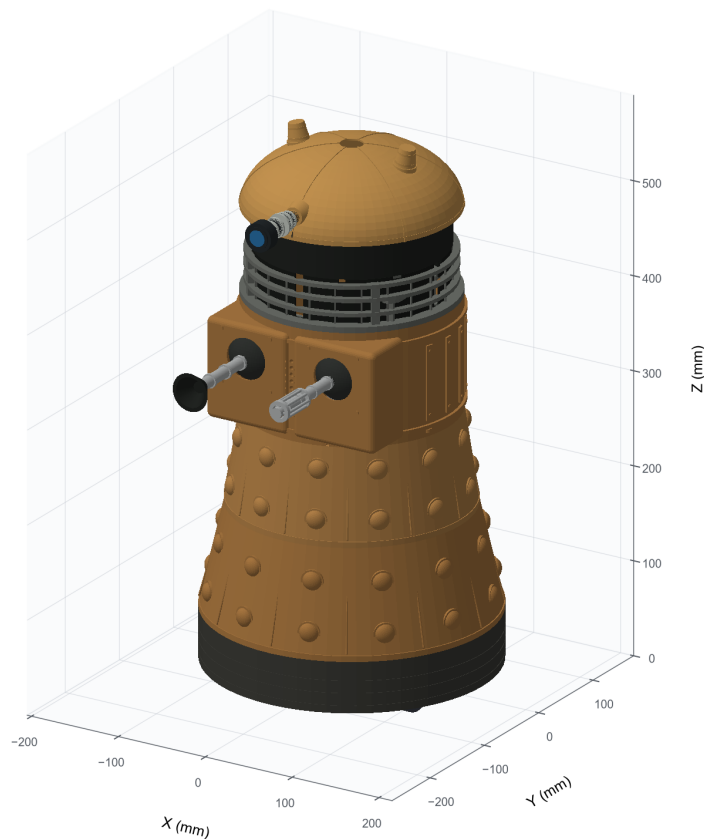
Nine STL designs form a reinforced circular motor base, a one-piece taller skirt, body sections that stack vertically, concealed arm mechanisms and an adjustable head friction drive. The base fits the Bambu Lab H2D in one print.

The robot keeps four TT drive motors, a rear ESP32 display, Wi-Fi access point and network controls, circular arm motion, a fifth TT motor driving the head through a friction wheel, and 24 original MP3 voices. The package includes editable CAD, circuits, mounting hardware lists and assembly instructions.

This is a digitally checked design for a first prototype. Physical fit, loaded traction, supply temperature and stopping behavior require the tests in this manual. Do not skip those gates.

Dalek ROUND-9 / assembly

Actual STL geometry | Purchased hardware uses nominal envelopes



Actual ROUND-9 STL geometry | Dimensions in mm | +Y is rear; -Y is front

CAD assembly view. Purchased parts and internal features may be simplified in the preview.

Using this package

Read the assembly sequence first, then use the exact part-specific mechanical instructions. Read the firmware procedure before wiring or flashing. The electrical research chapter explains the power and pin choices. Full-size circuit sheets and complete bills of materials follow the text.

The source package is in JeremyProffittOrg/robots on main under dalek/. Each drawing and source filename in this PDF is relative to that directory. Original files and firmware:
<https://github.com/JeremyProffittOrg/robots/tree/main/dalek> .

Use the PDF bookmarks to jump between chapters. The final appendix lists parts and wires. The supplied digital verification record reports what was actually checked and what remains unverified.

Assembly and commissioning sequence

Build and test one subsystem at a time. The mechanical chapter gives the exact part filenames, fasteners, bearing fits and assembly order. The electrical chapter and circuit sheets give the exact wire connections. This chapter places those operations in a safe order and defines when each stage is complete. A check is not complete until it has been performed on the physical robot.

1. Inventory before printing

Read the whole manual and open the CAD preview. Inventory `bom/electronics.csv`, `bom/hardware.csv`, and `bom/printed-parts.csv`. This revision has nine STL files. Print the shared pitch carrier twice, giving ten printed pieces. The lower and upper skirts are now one 213 mm print. All body sections remain complete upright prints with integral decoration and mounts. Keep purchased screws, bearings, retainers and spacers in labelled bags.

Check the actual board: it must be the original TTGO T-Display ESP32 with the 1.14-inch screen. Inventory four MG92B servos, five TT3777 motors, five 3766 wheels, three DRV8833 boards and the specified battery. Four motor/wheel pairs propel the base; the fifth turns the head by friction. Obtain the exact wheels before accepting either fit. Check board and servo part numbers against the drawings.

Tools: a digital calliper, metric hex keys, small Phillips drivers, flush cutters, a deburring tool, soldering iron, solder, wire stripper, crimp tool for the selected connectors, heat-shrink tube, digital multimeter, a current-limited bench supply, and a scale. A contact temperature probe and an oscilloscope improve the power and noise tests. Use eye protection when cutting wire or drilling. A threaded fastener, bearing, motor, battery or connector is purchased hardware, not a printed substitute.

Done when every required purchased item is present or its absence is recorded, the board variant is confirmed, and the tools can measure voltage and current. This is an inventory gate. It does not require assembling or energising anything.

2. Prove fit with small prints

Print `07_pitch_carrier.stl` and `10_head_motor_carriage.stl` first. These are functional parts of the nine-file set. Use them to check the servo horn, TT case, sliding slots and small screw fits with the intended material and slicer profile. Then print `01_base.stl` and check the actual motors and wheels in its integrated pockets before continuing with the body. Check each subsequent part's screw holes, bearing seats and board channels as it comes off the printer. Clean the first-layer edge before concluding that a dimension is wrong. Do not force a bearing or crush a board into a tight print.

If a fit needs adjustment, change the relevant clearance parameter in the OpenSCAD source, export that part again, and test it again. Do not scale the whole model in the slicer. Record the material, nozzle, layer height and clearance that passed. A printer-specific compensation may be needed, but it must preserve the joint centres and part lengths.

Done when fasteners pass without splitting the part, nuts retain their flats, the motor cannot shift in its holder, servo bodies seat without force, and the screen is visible with the USB connector accessible. The rear holder must retain the PCB by its edges without pressing on the glass.

3. Print and label the remaining parts

Follow the orientation and quantity fields in `printed-parts.csv`. The body order is `01_base`, `02_skirt`, `04_shoulder`, `05_neck`, and `06_head`. The number 03 is retired; other component IDs stay unchanged. Print each complete section upright at 100% scale. The 213 mm skirt uses more of the build height while preserving the robot's overall height. The base, motor pockets, battery support and structural walls remain one continuous print.

Use the H2D single-nozzle 0.4 mm profile for the circular base. Keep its floor flat on the bed. The 300 mm diameter plus an 8 mm brim allowance needs 316 x 316 mm. Use the verified left-nozzle placement at X162.5,Y160, within X0..325 and Y0..320. The package's `scripts/slice_h2d.py` uses an isolated copy of the installed settings and records the actual model, support and generated brim bounds. Do not use the combined 350 mm bed width as a single-nozzle limit. Use the specified PETG profile for the base and structural parts. Enable the recorded automatic supports and inspect the sliced adhesion paths before printing; a requested brim width does not prove that the slicer generated a brim around a supported feature. Remove supports fully from nut traps, screw holes, grooves and bearing seats. Keep paint out of the stacking registers and moving fits.

The one-piece skirt's checked job uses about 1033.87 g of PLA including support and brim. One 1 kg spool is insufficient. Have at least 1.14 kg available for that job including a 10% reserve, and arrange a compatible filament change or sufficient spool capacity before starting. The BOM's 3 kg PLA supply allowance covers this job and the other cosmetic parts.

Weigh each group after support removal. Record total printed mass and the eventual fully assembled mass. Do not use slicer infill percentage as a direct mass estimate for thin shells: most of a thin shell may already be walls. The validation report's solid mesh volume is a useful conservative bound for fully filled material, not a slicer timing estimate.

Done when every manifest quantity is made, identified, free from cracks and checked against its mating piece. Do not close the cosmetic shell yet.

4. Assemble the chassis without power

Follow the base and motor-retention steps in `mechanical1.md`. Inspect the continuous perimeter, floor, ribs and pocket walls. There are no base-quarter joints or suspended motor pillars. Fit the specified purchased retainers and washers. Tighten until the motor is secure; further tightening can crush its plastic gearbox or the print. Do not put liquid thread-locker on printed plastic.

Seat all four motors in the specified direction. Give their thin motor leads strain relief. Press each wheel onto its TT output shaft while supporting the gearbox body. The wheel's hub fit provides retention; do not drill the motor shaft. Check that tires cannot rub the bumper, a bolt, or a motor cable at any rotation. Check that the assembled chassis does not rock on a flat surface.

Thread the battery straps but leave the battery out. Seat and fasten the complete skirt on the bare base using the specified long driver and offset nut tool before fitting electronics. Prepare the FR4 plates using the actual PCB hole templates and the mechanical chapter's hardware keepouts. Retain the PCB nuts and washers below each plate, projecting no more than 3 mm; keep boards and upper spacers off during insertion. Lower the plate while centred at X0,Y0, slide it at base-local underside Z44, then lower it onto its spacers. Fit boards from above using the specified fasteners and measured screw-tip clearance, then lower the disconnected 70 x 114 x 76 mm battery into its pocket. Keep upper sections removable during commissioning.

Done when all wheels rotate by hand without body contact, motors cannot slide, the chassis rests evenly, and the battery can be removed without cutting wires. The mechanical chapter defines the exact floor and wheel clearances.

Before installing electronics and motors, perform the base proof-load check. Support the four motor-pocket floors on equal-height rigid blocks. Apply a distributed load equal to twice the intended complete robot mass to the actual battery and body-stack load areas for ten minutes. For a 4 kg finished robot, the proof load is 8 kg. Use the actual planned mass including hardware; the earlier 3 kg target is not a tested load rating. Keep all motors, wheels and the battery out of this test. Inspect before loading and after unloading. Fail the base if it cracks, a layer separates, a fastener boss loosens, or measurable permanent distortion prevents a flat fit or free motor insertion. This is a required physical builder test, not a strength result measured during design. A static proof test does not establish impact or fatigue life.

5. Build the power harness on the bench

Make the harness from the circuit sheets and `electronics/wiring.csv`, using the listed wire gauges and connector ratings. Put the main fuse close to the battery connector. Put the physical actuator switch in the feed to both movement converters. Keep the logic branch separately powered from the main switch. Terminate returns at the common ground point. Do not use a solderless breadboard or loose Dupont contacts for motor, servo or battery current.

Label every connector at both ends. Use a different keyed connector, or a clearly different connector size, for battery voltage and 5 V. Insulate exposed terminals and add strain relief. Confirm electrolytic capacitor polarity. Fit the motor noise capacitors close to the motor terminals. Never place a diode directly across a bidirectional motor as though it were a relay coil.

With the battery disconnected, check continuity of every wire against the net list. Check for shorts between each supply and ground. Inspect the physical switch and fuse ratings; an AC-only current rating is not a sufficient DC switch rating. Check the ADC resistor values with the meter before connecting an ESP32 input.

Done when every net has been checked, no supply is shorted, the battery connector polarity is correct, and the actuator switch interrupts both movement branches. Keep all downstream boards disconnected for the next stage.

6. Check the three supplies before connecting boards

Energise the harness from a current-limited bench supply at a voltage in the battery's operating range. Check each converter's unloaded output. Each 5 V branch must be within its specified tolerance and suitable for the attached devices. Then test each branch with an appropriate known load. Do not connect the battery voltage to a 5 V rail to make a failed converter test pass.

Confirm the actuator switch makes the motor and servo outputs fall to zero after stored capacitor charge has decayed. Confirm the logic output remains available. Check the external pack divider voltage against the calculation in the electrical chapter. At the maximum charge voltage it must remain below the ESP32 input limit. A reversed divider is a board-damaging error.

Done when the motor, servo and logic rails pass unloaded and loaded checks, the pack divider passes, and the physical actuator switch has the intended effect. A warm regulator alone does not prove a safe load; measure its temperature and voltage under load.

7. Flash and start the controller on USB only

Disconnect the battery and external controller 5 V supply. Leave all actuator supplies off. Connect the T-Display to the computer using USB. Do not back-feed the USB supply from an external 5 V source. Use the complete flashing procedure in `firmware.md`, including both the program and the LittleFS image. A program upload alone does not install the audio files.

After boot, check that the rear display is readable and shows network and safety status. Join the robot's access point using the credentials shown by the controller. Open the displayed local address. Check that the interface loads with the phone's Internet access disabled. It must display the sound list and controls without fetching a web font or script from a public website.

Store home Wi-Fi settings only through the local setup form. Use a 2.4 GHz network supported by the ESP32. Check the displayed station address after joining. If the home network is unavailable, follow the fallback procedure. Do not put Wi-Fi passwords in source files, a screenshot, this manual, or Git.

Done when the firmware and filesystem load, the TFT is readable, direct AP control works, station setup works on the intended network, and startup does not arm movement. Hardware-dependent status may correctly block arming until the next stages are complete.

8. Verify logic and hardware disable states

With USB removed, reconnect the external logic supply and the low-voltage control harness. Keep movement power off. Confirm the ground is common before connecting signal wires. Check the PCA9685 address and the buffer connections. Verify the motor sleep line is low and the servo output-enable line is in its disabled state at reset.

U8, the four arm-signal buffers, operates from 5V_SERVO. U9 routes head PWM and operates from 5V_MOTOR. Each signal and supply must match circuit sheet 03. GPIO2 has a 10k pull-down and supplies 20kHz head PWM. PCA channels 4/5 are static direction enables; never attach servos to these channels. Check that all three driver SLP pins and GPIO2 stay low during reset. No 5V power cable may connect to the ESP32's 3.3V pin. Compare the ADC display with a meter and apply its calibration factor if needed.

Done when reset leaves actuators disabled, the controller detects the actuator switch state, the PCA9685 responds, and battery voltage agrees with the meter to the stated tolerance. A blocked arm command is the correct outcome while power or sensor checks fail.

9. Test drive direction with wheels lifted

Support the chassis so all tires are clear of the bench. Keep hands away from the wheels. Connect the movement supplies and explicitly arm through the interface. Apply a brief, low command to both sides. Each wheel must turn in the direction that would move the robot forward. The right-side motors are physically mirrored, so their lead polarity must be checked, not copied by wire colour.

If one motor on a side turns backward, disarm, isolate power and reverse only that motor's two output leads. Do not connect two driver outputs together. Once directions are correct, test a slow forward command, a stop, a slow reverse command and a stop. Listen for rubbing and inspect driver temperature. Do not hold a stopped wheel against power to measure stall behavior.

Done when all four directions are correct, release stops the command, reversing passes through zero, and no tire or wire rubs. Leave final turning and loaded traction tests for the floor stage.

10. Centre servos before attaching the arms

Keep the arm horns and head drive connection detached. Command the positional channels to their documented centre pulse. Check that each servo stops and does not buzz against a hard limit. Fit the supplied horns at the neutral positions shown in the mechanical chapter. Retain each horn with its own centre screw. Do not print a replacement servo spline.

Assemble both gimbals while the shoulder is open. Feed the arm stem and decorative end outward through its socket from inside; the spherical root stays behind the opening. Insert each carrier through the central top opening,

lower it below the flange, then move it forward into its gunbox. Use the exact sequence and fasteners in the mechanical chapter. Fit the opaque black stretch-fabric socket liners with loose folds around the moving stems. Route each servo lead with a free loop and keep adhesive away from hinges, horns and wires. No servo case, horn or cable should be visible through a finished socket.

Start circular motion at 2 degrees and 0.10 Hz. Check one arm at a time. Correct a reversed axis using the calibration setting, not a cable swap on a three-wire servo. The revised maximum is 8 degrees; the older 12-degree limit does not apply to this concealed mechanism. The tip should trace a near-circle without contact, pulling a wire or tightening the fabric. Inspect the complete motion from front, sides and below to confirm concealment.

Done when both arms can complete repeated circles quietly, every fastener remains secure, no servo stalls, and the supply stays within tolerance. Continuous buzzing, a hot servo or a distorted orbit calls for a mechanical or calibration correction before proceeding.

11. Assemble and test the head

Bench assemble the two 608 bearings, spacer, shims, retainers and M8 spindle in the neck before seating it on the shoulder. Use the documented tool channels to fasten the neck. Fit the sliding head-motor carriage, fifth TT motor and horizontal 3766 wheel. Begin with the tyre retracted from the drum. Check every bearing seat and the eye-stalk sweep before fitting the dome.

With power isolated, fit and retain the head, then follow the jackscrew and guide-clamp sequence in `mechanical.md`. Increase contact only enough for reliable traction. The 3.6 mm slide range includes a released gap and a hard stop; it is not a calibrated force scale. Lock the setting, rotate the head through a complete turn by hand and check for binding. Use the specified access ports and short hex key. Do not tighten contact to force the head past a misaligned bearing or screw. Keep fixed motor leads clear of the rotating track.

Test low drive in each direction and zero output with the assembled robot supported. Record slip, coast angle, current and temperature. Done when the head turns through repeated full revolutions in both directions, zero removes powered drive, the tyre setting remains locked, and the head stays retained. Do not lift by the eye stalk. The controller sets motor drive level, not a measured head angle or speed.

12. Install and check audio

Fix the speaker to its body mount with the specified retainer. Keep both speaker terminals isolated from chassis ground. Wire the MAX98357A exactly as shown; its bridge speaker output has no grounded speaker lead. Confirm the gain resistor before applying power.

Play `hello.mp3` at low gain. Then test every sound button. Check for distortion, controller resets and motion pauses while audio plays. Keep gain within the speaker's 1 W limit and the logic supply budget. Warning phrases in the collection are available as sound effects; their filenames do not prove automatic event playback.

Done when all 24 files play on demand, speech is intelligible, sound does not reset the ESP32, and amplifier/speaker temperature stays acceptable during repeated use. Keep the speaker and amplifier wiring fixed in the body.

13. Run the stop tests before floor driving

Keep wheels lifted for this complete test. Arm and command slow motion. Release the drive button and confirm the command stops. Repeat and change browser tabs. Repeat and close the page. Repeat and disable the phone's Wi-Fi. Repeat and turn off the access point or station connection used for control. Measure the delay to removal of the drive signal; the firmware lease is 500 ms, and the exact task and I/O delay must be checked on hardware. Tire coasting after removal of power is not the same as a continued drive signal.

Repeat every stop case with the head rotating and arms moving. A zero head setting must remove PWM immediately. Measure at least 100ms with both head bridge inputs low on a direction reversal, including a quick move through zero. The rotating head can coast after drive ends; record its coast angle. With head motor disconnected, scope the two U11 inputs and confirm only one carries 20kHz PWM at a time, capped at 100/255 duty.

Turn the actuator switch off during a slow command. Wheel, head-motor and arm-servo power must be removed. Turn it on again. Movement must not resume until the operator explicitly arms again. Reset the ESP32 during a slow command and verify no restart. Check that a second browser cannot revive a stale command after a stop.

Test low-battery detection with a bench supply, not by repeatedly exhausting the actual battery. Follow the voltage thresholds in the electrical and firmware chapters. Restore normal voltage and confirm recovery still requires explicit arming. Temporarily disconnect the servo-controller communication while safely supported, and confirm the documented fault response. Reconnect with power isolated.

Done only when every stop case passes. Record the measured time and test setup. Do not move to floor testing with any unexpected restart or uncontrolled motion.

14. Close the shell and run a short floor test

Lower each complete body section onto the section below it, using the mechanical chapter's assembly sequence. The register locates the joint; fasteners retain it. Check that each register seats fully before tightening. The five body pieces are the base, combined skirt, shoulder, neck and head. Check rear-screen, switch and service access. Route fixed cables clear of wheels, arms, the head track and fastener tips. Leave service loops so complete sections can be lifted apart. Tighten the battery straps and confirm the pack cannot shift.

Weigh the complete robot. Begin on a clear, level hard floor with an operator beside the physical stop switch. Use low speed and broad forward arcs. Do not begin with a stationary pivot, carpet, ramp or threshold. The specified tires can create more steering resistance than these small motors can overcome. Stop immediately if a wheel stops turning while commanded.

Run for one minute, isolate power and inspect joints, cables, tires, motor temperature and converter temperature. Repeat for five minutes, then fifteen minutes if the earlier test passes. Record battery voltage, mass, floor type, speed setting, current and temperatures. A conservative commissioning gate is to stop testing at 50 C measured on a motor, servo or nearby printed mount, or on any sign of softening, odour or repeated electrical shutdown. This is a build-test threshold, not a replacement for manufacturer component ratings.

Done when the fully dressed robot drives, steers in moving arcs, moves both arms, turns the head, plays audio and stops correctly under the same measured load. This is the first physical proof of the design's operating envelope.

15. Charge, maintain and store

Switch off and disconnect the robot from the battery before charging. Use the specified Bioenno charger on the pack's charging connector. Do not charge through the ESP32, an Adafruit single-cell charger, or a converter output. Charge on a stable nonflammable surface with ventilation. Follow the battery manufacturer's limits and instructions.

Before each use, check the straps, wheel fit, arm horns, head retention and exposed wires. Test the physical stop switch and release-to-stop behavior. Stop use if the battery is damaged or swollen, a connector is discoloured, a motor binds, or a printed bracket cracks. Replace a damaged structural part rather than gluing across a loaded fracture.

Do not carry the robot by an arm, the eye, dome decoration or upper body section. Lift from the single-piece base with battery power isolated. Keep the wheels free of hair and thread. Store the battery disconnected according to its manufacturer's guidance. Keep a copy of the calibrated firmware settings with the robot.

Fault guide

- Screen works but movement is blocked: check actuator switch state, pack ADC calibration, PCA9685 response, explicit arming and command timeout. Read the displayed fault before changing code.
- Controller resets when a motor starts: check rail sag, common ground, motor capacitors, converter load and connector resistance. Do not disable brownout protection.
- One wheel turns backward: isolate power and swap that motor's two bridge leads. Check all four wheels again while lifted.
- Arms buzz or fail to circle: check horn centres, direction calibration, gimbal contact, servo supply and 5 V signal buffers. Reduce amplitude while diagnosing the fit.
- Head moves under power at zero: turn ACTUATORS off. Check GPIO2 is low, the two U9 direction gates and U11 input wiring. There is no head-servo neutral adjustment. Distinguish a short coast from continued powered motion. Binding at one angle indicates track alignment, fastener intrusion or dome clearance; do not increase tyre pressure to overcome it.
- Robot moves straight but will not pivot: use moving arcs and a lower-friction floor. Measure mass and current. Do not raise motor voltage or remove current limiting.
- Audio is missing: confirm both firmware and LittleFS were flashed, inspect the sound list, verify I2S pins, and check speaker wiring and amplifier gain.
- Home network fails: join the local AP and follow the firmware recovery instructions. Reconnection must not restart motion.

Physical test record

Record date, builder, firmware commit, slicer profile, material, loaded mass and floor type. For each stage above, write PASS or FAIL with the measured result. Include supply voltages under simultaneous load, pack-voltage calibration error, stop delay, fifteen-minute temperatures and observed runtime. Until that record exists, the package remains a digitally checked design for a first prototype.

ROUND-9 mechanical design and assembly

The revised robot has a circular 300 mm base, four concealed ground wheels, complete vertically stacked body sections, concealed arm servos, and an adjustable TT-motor friction drive for the head. There are nine STL designs and ten printed pieces: print `07_pitch_carrier.stl` twice. Each STL is one connected part. Nominal assembled height remains 563.8 mm, below the requested 914.4 mm limit. One 213 mm skirt replaces the two former skirt prints and removes their bolted middle joint. The shoulder remains removable for service. Fewer joints may improve rigidity; no strength improvement has been physically measured.

The reference review covers 100 distinct photographs. The resulting details include a dark round fender, four rows of hemispheres with raised bezels, twelve aligned skirt seams, shoulder slats and rivets, rounded gunboxes and spherical arm roots, three neck rings over a dark liner, eyestalk discs, a stepped lens, tilted ribbed lamps, dome seams, telescoping arm bands and an eight-rod decorative emitter. The circular footprint and hidden motor drive are custom adaptations. See `appearance-research.md` and `reference-image-review.csv`.

This is a digitally checked first-prototype design. No print strength, motor payload, tire grip, runtime, actual bearing fit or assembled robot motion has been physically measured. Mesh checks, component-envelope sampling and real H2D slicing establish the limited digital results described in `revision-review.md` and the generated validation reports. Complete the physical gates below before operation.

Print inventory

Use the released STL files, a 0.4 mm nozzle, 0.20 mm layers and single-nozzle H2D mode. The single-nozzle envelope is 325 x 320 x 325 mm. The circular base plus its requested 8 mm brim is 316 x 316 mm. The other parts request 5 mm brims. Keep the print on the bed. The actual slicing report is authoritative for generated supports, brim paths and the exact released STL hashes. The checked process uses normal automatic supports. Do not substitute a different orientation or auto-arrangement without checking its complete toolpath envelope. [Bambu H2D specification](#).

1. `01_base.stl`: PETG, floor down. Six walls, six top/bottom layers and 40% gyroid. Remove supports through the bottom wheel openings. The body is diameter 300 x 54 mm, plus a 3 mm registry tongue. Its floor is 6 mm thick. Main ribs are 4 mm thick. The outside bumper remains continuous around the full circle.
2. `02_skirt.stl`: PLA, large opening down. Four walls and 15% infill. One complete body changes from diameter 300 to 250 over its first 110 mm, then to diameter 220 over the next 100 mm. The 3 mm top tongue gives a total print height of 213 mm. The four hemisphere rows and exterior relief are retained. An integral tapered rib around Z110 replaces the middle flanges and bolts; its minimum clear bore is 240 mm. The limiting top opening is 184 mm diameter. The 213 mm print fits below the requested 320 mm height ceiling, which leaves 5 mm below the H2D maximum. This change combines sections without stretching the robot.
3. `04_shoulder.stl`: PETG, bottom flange down. Four walls and 25% infill. Its body is 120 mm high, plus its tongue. Both concealed gunboxes, servo pockets, speaker baffle and rear display frame are integral. Remove support from inside each gunbox before installing hardware.
4. `05_neck.stl`: PETG, bottom flange down. Four walls and 30% infill. Rings, opaque grille liner, bearing tower, slide deck, adjustment lug, hard stop and service ports are integral. Clear the bearing seats, lower shaft slot and all tool channels.
5. `06_head.stl`: PLA, drum rim down. Four walls and 15% infill. The friction drum, hub, spokes, dome, eye and lamps are one part. Remove internal support through the drum and crown openings. Keep the inside drum surface smooth and free of paint, glue and support scars.

6. `07_pitch_carrier.stl`: two identical PETG parts, floor down. Four walls and 30% infill. Each carrier is 49 x 30 x 49 mm. Its Y bounds are -10 to +20 mm, not symmetric about the yaw shaft.
7. `08_plunger_arm.stl`: PLA, as exported and placed on the bed. Four walls and 15% infill. The spherical cover, stem bands and cup are integral.
8. `09_emitter_arm.stl`: PLA, as exported and placed on the bed. Four walls and 15% infill. The spherical cover, eight rods and muzzle are integral. This is decorative and has no projectile mechanism.
9. `10_head_motor_carriage.stl`: PETG, floor down. Four walls and 30% infill. Its radial slots and padded motor saddle form one part. Existing part identifiers are retained; there is no separate `03` print.

Paint the bumper, neck liner, outside of the head drum, socket roots and plunger cup near-black. Use bronze body panels, gold relief, silver stems and pale eye discs. Paint the outside only; keep all bearing seats, sliding faces, horn seats and the friction track clean. Read actual mesh dimensions and material-volume bounds from `bom/printed-parts.csv`. Read predicted deposited mass and support consumption from the current H2D slice report. Neither number is a verified assembled mass.

The actual combined-skirt slice predicts 1,033.87 g of PLA including supports and brim, and about 23 hours 46 minutes of printing. A single 1 kg spool is insufficient. Have at least 1.14 kg available for this print with a 10 percent reserve. Arrange a compatible filament change or a sufficient continuous supply before starting. Verify that procedure in the actual printer setup. The three-kilogram PLA BOM allowance also covers other cosmetic prints and a large-print retry contingency. The predicted installed skirt material is 527.879 g after removing supports and brim.

Coordinate map and stack joints

Dimensions are millimetres. Positive Y is rear, positive X is right and Z points up. World Z0 is the floor. The nominal base datum is world Z13.8: wheel radius 31.5 minus floor thickness 6 minus the drawing-derived 11.7 mm case-bottom-to-axle datum. Actual motor padding and axle height must be measured.

- Base origin: world Z13.8; body top 67.8.
- One-piece skirt origin: 67.8; slope transition 177.8; body top 277.8; tongue top 280.8.
- Shoulder origin: 277.8; body top 397.8.
- Neck origin: 397.8. Lower bearing is local Z40-47; upper bearing is Z59-66.
- Head STL origin: world Z449.8, or neck-local Z52. Its drum starts at print Z0. Its hub foot is print Z15, world Z464.8. Dome rim is world Z488.8. Lamp tops are approximately 563.8.
- Ground-wheel axes: X +/-98, Y +/-58, world Z31.5. The fully seated hub accommodation range is $\text{abs}(X)$ 95.5-100.5, not an instruction to stop a hub short.
- Arm yaw axes: shoulder-local (-47,-94,42) and (53,-94,42). Each pitch-axis offset in the yaw frame is (-3,0,35). Neutral spherical centers are world (-50,-94,354.8) and (50,-94,354.8).
- Head carriage origin at nominal contact: neck-local (64.5,0,29.2). Head wheel center: (64.5,0,71); its shaft is vertical and its tire occupies local Z56.5-85.5.

Four M4x20 screws retain each of the three horizontal body joints, twelve screws in total. Hole radii are 137 mm for base/skirt and 97 for skirt/shoulder and shoulder/neck. All four holes are at 0, 90, 180 and 270 degrees. Remaining flanges are 18 mm wide and 6 mm thick. The tongue is 2 mm wide and 3 mm high; its groove provides nominal 0.3 mm clearance per side. Align the hemisphere columns and the front seam before tightening. The tongue locates the joint; bolts retain it. The former middle joint, its four screws, four nuts and eight washers are removed.

Use a 250 mm-shaft ball-end 3 mm driver rated for at least 20 degrees for the base/skirt joint. The sampled approach tilts the shaft 16 degrees toward the center from each bolt at skirt-local radius 137, Z10. Its upper shaft clears the 184 mm top opening. A 150 mm driver is too short for the combined skirt. Use the long-reach 7 mm nut tool listed in the hardware BOM below the base flange; dry-fit the actual tool before final assembly. The neck provides four 9.2 mm vertical tool channels, so its bolts can be tightened with a straight driver before fitting the head.

A documented driver option is [Stahlwille 10507N KK](#), [article 73103003](#): 3 mm hex, 300 mm blade and a listed 25-degree swivel angle. Its long bare blade meets the minimum reach. Keep large bit holders or couplers out of the shared space beside the crowfoot's universal joint; the nominal checks use the bare hex shaft there.

The specified deep nut tool uses a [Bahco 677-7 crowfoot](#), [6966 universal joint](#) and [6960-11 extension](#). The 7 mm crowfoot has 5 mm-thick jaws, 22 mm width and 17.5 mm between the nut and drive centers. The universal joint is 35 mm long and 13.9 mm diameter. Use an extension at least 250 mm long. The crowfoot holds the nut while the 3 mm ball-end driver turns the screw; its offset drive passes inside the base flange. First lower the tool with its nut axis at radius 80 mm and its jaws at base-local Z42-47; slide it outward onto the nut at radius 137. Its drive axis is radius 119.5. Tilt the extension 16 degrees inward above base-local Z82. Keep the battery and plates out during this operation. Check the actual tool's unlabelled jaw projection and drive-boss height against the bounded envelopes below before relying on this path. Their reproducible samples are in `scripts/check_skirt_access.py`, with results in `cad/skirt-access-checks.json`.

At each bolt, the checked maximum tool envelopes use a radial axis pointing outward and a tangential axis across the tool. The thin jaws occupy radial coordinates 128-143 mm, tangential +/-11 mm and base-local Z42-47. The thicker drive boss occupies radial 112.5-128, tangential +/-7 and Z42-51. The universal joint stays within 14 mm diameter about radius 119.5 from Z47 to Z82. Keep the tilted extension and its couplers within that 14 mm diameter. Rotate this same envelope around the base for the other three bolts. These are acceptance limits for the actual tool, not claims that every unlabelled dimension of the listed crowfoot was supplied by its manufacturer. Hold the crowfoot steady; a rotational sweep under the flange was not checked.

Verify purchased parts before printing

The drive uses five Adafruit 3777 TT motors and five Adafruit 3766 wheels: four pairs drive the base and one drives the head. The wheel is nominally 63 mm diameter by 29 mm wide. The motor drawing identifies an 18.6 mm gearbox, 22.4 mm metal-end width, 22.44 mm case height, 70 mm length and 36.6 mm across shaft ends. The product page's simplified body dimensions differ from that drawing; the CAD uses the drawing envelopes. Measure the real shaft, fully seated hub, case and leads. [Motor](#), [drawing](#), [wheel](#).

The arm motors are four MG92B servos. Adafruit specifies a 36 x 12 x 31 mm overall envelope including mounting features. The CAD provides measured-fit padded mounts; the modeled 24 mm case length is an assembly envelope, not a claimed manufacturer case drawing. Check the actual body, mounting ears, spline location and supplied horn. Use the supplied spline horns and center screws. Do not print a replacement spline. [MG92B](#).

The original TTGO T-Display PCB is nominally 51.52 x 25.04 mm. The rear recess is 52.2 x 26 mm. Insert the confirmed original 1.14-inch ESP32 board from inside, behind the lip. Screen, buttons and USB remain accessible. The S3 and AMOLED boards are different. [Original board files](#).

The Adafruit 1313 speaker is 77.8 x 77.8 x 25.49 mm with 60 mm mounting centers. Its new internal baffle is at Y-20 to -16, center shoulder-local Z43. The speaker body extends rearward. The battery is the specified Bioenno BLF-1206A 12 V, 6 Ah pack. It is rotated in plan to 70 mm X by 114 mm Y, with nominal height 76 mm. Use the matching charger listed in the electrical BOM. [Speaker drawing](#).

Build the circular motor base

1. Clear supports, shaft ports, clamp holes, wiring exits and strap slots. Inspect floor and ribs for voids, warping or cracks. Confirm all four motor floors are coplanar. Do not scale the whole part to correct one hole.
2. Press each real wheel fully onto its motor outside the base. Motor centerlines are $X \pm 70.5$, with axles at $Y \pm 58$. Motor tails point toward $Y0$ and stop nominally at $Y \pm 1$. Leads leave upward through the central openings. Do not put wire slack between the motor ends.
3. Lower the four motor/wheel pairs from above. Cases sit on the 6 mm floor. Add measured side pads for the narrower gearbox. Under-case pads change axle height and must be applied consistently if required. Both shaft ends must turn freely.
4. Use two stainless strips per motor, each $43 \times 8 \times 1$ mm, with 3.4 mm holes 37 mm apart. Deburr and place nonconductive padding under each strip. The mirrored clamp axes are $\text{abs}(X)$ 51.5 and 88.5 at $\text{abs}(Y)$ 6 and 16. The bosses end at local Z29. Adjust padding so the strip lies flat without crushing the case.
5. Install sixteen M3x40 screws, washers and nuts. The full motor floor carries the cases; side saddles resist lateral motion and metal strips provide retention. Check screw tips retain ground clearance. Turn every wheel completely and inspect the hub, tire runout and both axle ends.
6. The wheel opening is limited to radius 147, retaining a continuous 3 mm outer bumper at its closest location. Its inboard boundary is $\text{abs}(X)$ 80.5 and its roof ends at local Z52. The nominal tire top is local Z49.2. At the most inward accommodated hub, the tire's inner face is X81. The electronics plate ends at X80; that 1 mm nominal gap needs a real loaded-runout check. Never leave a wheel partly seated to obtain it.
7. Keep all motor leads, suppression capacitors and strain relief above the well and away from the rotating tire. Reject any rubbing before proceeding.

Battery, raised plates and lower stack

Bolt the complete skirt onto the bare base before installing the electronics plates or battery. Fit the nuts below the upper base flange and use the 250 mm angled driver through the skirt. Seat the tongue completely before tightening. The top opening has a 92 mm radius. Keep the battery out until both plates and their boards are fitted.

Prefit the two battery straps through the empty pocket before closing the base. The battery pocket has 76×116 mm clear plan space and a continuous 6 mm floor. After the plates and boards are fitted, lower the $70 \times 114 \times 76$ mm pack centered through the top opening into its pocket. Its half-diagonal is approximately 66.9 mm, below the 92 mm opening radius. Use two independent 20 mm straps through the slots at $X \pm 34$, $Y \pm 35$. Pad and restrain the pack without crushing it. The left connector notch is X-46 to -36, Y-22 to 22. Route its lead upward in the fixed side gap; secure the lead before it can reach a motor tail.

Cut two unclad FR4 plates to $160 \times 56 \times 2$ mm. Round and deburr the edges. Their centers are (0,-93) and (0,93), so X is -80 to 80 and Y is -121 to -65 or 65 to 121. Eight plate mounts are $X \pm 54$ at Y -93,-69,69,93. Before fitting the shell, insert M2.5x40 screws upward from below the base and place 25 mm nylon spacers on its 9 mm-high pads, placing final plate undersides at base-local Z34. Fit the plate washers and nuts from above after seating the plates, and check two complete threads beyond each nut. Support the unpowered chassis securely on blocks to reach the screw heads from below.

Prepare the underside PCB nuts before placing either plate in the robot. Use the actual PCB as the drilling template. Retain each underside washer and nut with small cured nonconductive epoxy side fillets. For the general M2.5 board mounts, use a 0.5 mm washer and thin M2.5 hex nut no more than 1.6 mm thick. The two Pololu regulators instead use the M2 hardware specified below. Use thin DIN 439 / ISO 4035 nuts at PCB locations that accept M2.5 screws; 32 sets are allocated including spares. Retain only the sets needed by the actual board holes.

Keep the eight standard nuts for the base-to-plate mounts. [Wuerth 032825 dimensions](#) confirm the 1.6 mm height and 5 mm wrench size. The adhesive retains position during assembly; the tightened screw and nut carry the board load. Keep adhesive off the threads, board and hole. The complete retained washer, nut and adhesive must project no more than 3 mm below the FR4 and stay inside its outline. Keep that hardware and adhesive wholly outside an 8 mm-radius keepout around every base-to-plate mount axis. Check actual PCB templates and hole positions against those keepouts before drilling or bonding. Unknown actual hole locations are a physical fit gate. Keep the specified electronics layout unless any changed placement has been checked for clearance. Align and test each thread on the bench.

Lower each plate with its retained underside hardware horizontally at X0, Y0 through the open skirt. Its centered half-diagonal is 84.76 mm, leaving nominal 7.24 mm radial clearance at the top. Lower the FR4 underside to base-local Z44, translate its center to Y-93 or +93, then lower the FR4 underside to Z34 onto the spacers. During the horizontal move, the retained hardware ends at or above Z41, giving at least 1 mm clearance above nominal 40 mm screw tips. The plate top is Z46, 2 mm below the base flange underside. Keep the battery and tall components out of this path. Fit and wire the boards after the prepared plates are seated. Do not lower a populated plate straight down at its final outer position through the smaller top opening.

Use a 5 mm socket with the same universal joint and long extension for the M2.5 nuts. Keep the socket/joint envelope within 14 mm diameter. Start with the tool axis at radius 75 mm, lower the socket bottom to just above base-local Z45, move outward to the target nut, then lower onto it. Tilt the extension 8 degrees inward above base-local Z82 to clear the top bore. The outer plate mounts are radius 107.5 mm and the inner mounts are radius 87.62 mm. Retain access to the below-base screw heads while tightening. Install the board fasteners while the battery and adjacent tall components are absent; dry-fit the actual driver and nut tool before closing the shoulder.

Use each actual PCB as a drilling template on the FR4; do not drill or assume a universal PCB pattern. The two 81 x 51 mm protoboards lie with their 81 mm dimension along X, centered at X-35, Y +/-93. Use the unchanged 6 mm PCB spacers. Insert M2.5x14 board screws from above through two 0.5 mm head washers into the pre-retained underside nuts; one 0.5 mm washer is below the FR4. If trimming a 16 mm screw, measure the final working length and deburr its end before assembly. Check at least two complete exposed threads and at least 2 mm clearance from every motor case. The thin nuts increase the nominal permitted screw-length interval, but actual washer, nut, PCB and motor-padding dimensions still control acceptance. With a 1.6 mm PCB, 1 mm total head washers and a 14 mm screw, the nominal tip is base-local Z30.6, giving 2.16 mm above the drawing-derived motor-case top. Do not substitute thicker PCB nuts. Do not close the shoulder if both limits cannot be met. The retained nuts remove the need to insert board screws or hold loose nuts below an installed plate. The Pololu 4091 regulator is an explicit exception: [Pololu specifies three 0.086-inch mounting holes for M2 screws](#). Use three M2x14 screws per regulator, six total, with matching M2 nuts no more than 1.6 mm thick. Use two 0.3 mm washers under each M2 screwhead and one below the FR4. The existing 6 mm spacers may be used only if their clearance bores pass the M2 screw freely and their ends support the actual board pads. Do not force M2.5 screws into those holes or enlarge the regulator PCB holes. Pre-retain the M2 underside nuts by the same method. Determine and trim the actual M2 working screw length from its own washer and board stack; with an actual 1.6 mm PCB and the specified two 0.3 mm head washers, a 13.5 mm working length gives a nominal tip at base-local Z30.7. Do not assume an untrimmed 14 mm screw meets the clearance requirement. Require at least two full exposed M2 threads and 2 mm of motor-case clearance. Other boards use fasteners matched to their actual holes and templates. Component positions and connector clearance are specified in the electrical chapter. The third DRV8833 drives the head. It fits on the existing rear plate; there is no extra printed electronics mount.

Test disconnectable harness loops before closing the removable shoulder joint. For plate service, remove the shoulder and battery, disconnect and remove tall boards where needed, raise each plate with its retained underside nuts to base-local Z44, move it to the center, and lift it through the skirt. Remove the plates before accessing motor clamps or base-joint nuts. The deeper shell needs the listed long tools; it cannot be separated at its middle for service.

Concealed arm mechanisms and rear board

Assemble the shoulder on the bench with its top opening accessible. Both MG92B pairs and their horns belong inside the rounded gunboxes. Only spherical roots and clean stems emerge from the sockets. The final software limit is 8 degrees on each arm axis; do not restore the former 12-degree limit.

1. Center the four servos electrically before attaching their supplied horns. Keep the arms detached. The yaw support face is shoulder-local Z8 and the target horn face is Z42. Select pads from actual measurements. Thread two cable ties through each yaw pocket before installing the servo; tighten without crushing the case.
2. Keep both pitch-servo carriers outside the shoulder while feeding the arms. Feed each detached arm tip from inside through the 44 mm front socket. The 40 mm plunger cup is the limiting part. Check the actual painted cup with a caliper. The spherical cap stays inside; it does not pass through the smaller front opening. An installed pitch servo blocks this feed path.
3. A nominal insertion path tilts the detached arm 20 degrees downward and keeps its axis through socket center Y-115, Z77. With X at +50 or -50, move the arm pivot from Y+30 to -20 while $Z = 77 + (Y + 115) * \tan(20 \text{ degrees})$. Keep Y-20 while reducing tilt from 20 to zero and keep $Z = 77 + 95 * \tan(\text{tilt})$. Then advance the horizontal arm pivot from Y-20 to -94 at Z77. Keep the stem centered in the opening. Support both arms in this final horizontal position with padded blocks under the exposed stems. Their root centers are shoulder-local (-50,-94,77) and (50,-94,77). Keep the blocks outside the body and leave the interior clear. Do not let the unsupported arms load or scrape the socket edges.
4. Assemble each pitch-servo carrier outside the shoulder. The carrier's yaw-horn attachment is its floor. Attach the pitch servo to the carrier's side wall with its supplied horn facing the pitch axis. Use the two tie locations and a measured pad. Relative to the yaw origin, the checked pitch-case envelope is X4..31,Y-6..6,Z13..49. The pitch axis is (-3,0,35). Keep the lead loop behind the carrier, with slack for both axes.
5. Keep both arms supported while inserting each carrier and pitch servo. Pass the assembly through the central top opening at approximately X+29 or -29, Y-44. The right carrier's extreme plan corner is (64,-54), radius 83.74, inside the 92 mm opening. Lower the carrier floor to Z46. First move it sideways to its final yaw X53 or -47 while keeping Y-44. Then move it forward to Y-94 and lower it to Z42. Seat it on the centered yaw horn and install the supplied horn-center screw and the two M2 fasteners. Do not force the assembled gimbal straight down at its final forward position. This path clears the supported arms in the nominal geometry.
6. Attach each supported arm plate to its supplied pitch horn using two M2x10 screws, washers and nuts, plus the horn's supplied center screw. Remove its support only after the horn and fasteners retain the arm. Both arm STLs include a 28 mm print-coordinate offset. The assembly transform removes that offset, rotates 90 degrees about X and -90 degrees about Z, then applies pitch and parent yaw.
7. Fit an opaque flexible socket liner to each gunbox so no motor, horn, tie or lead is visible around the moving root. Cut a 44 mm central opening in the listed black stretch fabric. Use a radial slit with an overlapped seam. Bond the inner edge inside the rear rim of the spherical cap and the outer edge around the fixed socket. Retain loose folds between these attachment edges. Keep adhesive on fixed surfaces and the intended moving attachment only. Do not allow fabric to enter a horn, bind an axis, contact the speaker cone or pull a lead. The opaque liner is purchased fabric, not an extra STL. Inspect from the front, both sides and below at all four yaw/pitch extremes.
8. Install the speaker on its internal baffle and put acoustic cloth behind the front grille. Insert the T-Display from inside the rear frame. Use four padded retaining tabs, M3x25 screws, washers and nuts. Touch only bare PCB edges with the pads; keep components, antenna, display glass, USB and both buttons clear. Fit the two rear switches and service plugs.

Run twenty slow circles at a small amplitude before increasing to 8 degrees. Reject buzzing, wire tension, fabric snagging or a visible actuator. The spherical cap has radius 28 with a 2.5 mm wall, while the fixed spherical cavity is radius 29.2. These are nominal CAD clearances, not permission to force an oversized painted print through the socket.

Bearing stack and adjustable friction drive

The head load travels through the hub, steel spindle, two 608 bearings, metal inner-race spacer and neck structure. The tire supplies tangential drive and adjustable contact; it does not support the head's weight. There is no head servo, belt or servo pulley.

1. With the neck still on the bench, clean both 22.2 mm bearing seats. Fit the lower 608 bearing at neck-local Z40-47 and upper bearing at 59-66. Put the 12 mm-long, 8.1 mm-ID, 12 mm-OD metal spacer between their inner races. Fit the narrow 8 x 12 x 1 mm shims below and above the inner-race stack. Retain outer races with three M3 plastic thread-forming screws and 9 mm washers at each end. Do this before fastening the neck onto the shoulder; its lower retainer is not inserted through a service port.
2. Insert the M8x70 spindle from below. It must pass freely through both 8 mm bearing bores. Keep the narrow shims on the inner races and verify free rotation. Do not pull a misaligned bearing into place with the spindle nut.
3. Lower the neck onto the shoulder, connect the fixed harness and tighten its four M4 joints through the vertical tool channels. Leave the head off.
4. Fit the fifth TT motor on its side in the head carriage. Its axle is vertical; the metal tail points toward negative Y. Use measured thin pads and two ties through the saddle. The nominal motor transform from the original drawing frame (x,y,z) is neck-local (64.5+z-11.7,y,43.5+x). The metal-end bottom is Z32.3 and the carriage floor top is 32.2. Support the narrower gearbox with measured padding rather than forcing it down. The unused lower shaft has a 15 mm-wide radial clearance slot through both carriage and fixed deck.
5. Fit its wheel fully. Nominal wheel center is neck-local (64.5,0,71). Measure actual seating; use appropriate case-height shims to keep the 29 mm tire within Z56.5-85.5 and clear of the head spokes at Z91. Both shaft ends and the hub must remain free. Keep motor wires fixed below the rotating drum.
6. Install the carriage on the fixed deck using four M3x16 screws. Fixed screw axes are (43,-50),(43,8),(83.5,-40),(83.5,8). Use 6 mm-OD, 1 mm-thick washers above the carriage and standard washers and M3 nyloc nuts below the deck. The small top washer diameter is required for the full slide travel. Leave the screws just loose enough to slide; retain all nuts. The deck is Z25.2-29.2; carriage floor is 29.2-32.2.
7. Insert one M3 nut into the open-top adjustment-lug pocket. Fit the M3x25 jackscrew along positive X through Y-35, Z38. The screw tip bears on the carriage pad. Start with carriage X61.5, nominally 3 mm clear of the inside track. Radial slots permit movement to X65.1; the fixed stop at X90.1 limits the outward plate edge. Do not enlarge slots or bypass that stop to cure a binding head.
8. Lower the head over the spindle with the tire released. The drum is 192 mm inside diameter and 198 mm outside diameter. Its bottom is neck-local Z52 and the spokes start at Z91. Fit the M8 washer above the printed hub and the nyloc nut using a 13 mm socket no more than 23 mm outside diameter through the 24 mm crown hole. Set retention while keeping free bearing rotation. Do not tighten until the head binds.
9. Turn the dome by hand through two complete revolutions while the tire remains released. Check drum roundness, axial runout, crown retention, eye sweep and clearance from the stationary liner. The liner ends at Z51, giving nominal 1 mm clearance below the rotating drum.
10. With power off and the head seated, reach the jackscrew from the left through its service port at Y-40..-30,Z34.5..41.5. Advance it only until the tire makes light, reliable contact. Nominal geometric contact is

X64.5. The final 0.6 mm travel is a geometric allowance, not a measured force setting or a required amount of compression. Excess pressure loads the motor shaft and head bearings.

11. Hold the adjustment while tightening all four slide clamps against their M3 nyloc nuts. These clamps lock the setting; there is no separate jackscrew jam nut. The clamp screws remain accessible with a 2.5 mm L-key whose short leg is 15 mm. Its horizontal shaft runs near neck-local Z49.9 through the 3.8 mm-high ports; the short leg reaches the screw socket near Z34.9. Approach the two negative-Y screws from the front and the two positive-Y screws from the rear at their specified X coordinates. Use a correctly sized tool; do not pry open a printed port.
12. Recheck two full hand turns after locking. Start powered tests at low duty in each direction. Increase contact only enough to avoid unwanted slip, then lock and repeat. Do not increase pressure to overcome an eccentric drum, tight spindle nut or snagged liner. Recheck after the motor and PETG mount warm during the short commissioning run.

The head controller commands speed and direction; it has no measured angle feedback. The rolling ratio is approximately $96/31.5=3.05$ motor-wheel turns per head turn when there is no slip. Actual slip, speed and stopping distance must be measured. The eye and dome lights are decorative and contain no moving wires, so no slip ring is required.

Physical acceptance and service

- Before motor installation, support all four motor-floor regions on equal rigid blocks. Apply twice the intended complete robot mass across the battery and stack load points for ten minutes, close to the floor. Unload and inspect for cracks or permanent distortion. This is a proposed proof test, not a verified rating.
- Verify fully seated wheels and free rotation unloaded and under the intended mass. Check the minimum plate/tire gap, bumper clearance and ground clearance with actual tire deformation.
- Weigh the complete robot. Four wide tires resist tight turns. Start on a level hard floor with slow starts and broad moving arcs. Measure current and temperatures. Do not assume these small TT motors can pivot in place, climb thresholds or run on carpet because the base is reinforced.
- Run twenty arm circles while inspecting every socket from low, side and frontal angles. The fabric must remain opaque, loose and clear of moving hardware at all permitted poses.
- Run a full head turn in each direction, check runout and repeat all contact/lock checks warm. A slipping drive is preferable to forcing a bound bearing with excessive pressure.
- Complete the electrical stop, Wi-Fi-loss, power-switch and simultaneous-load tests in `assembly.md`. The physical actuator switch must remain reachable.

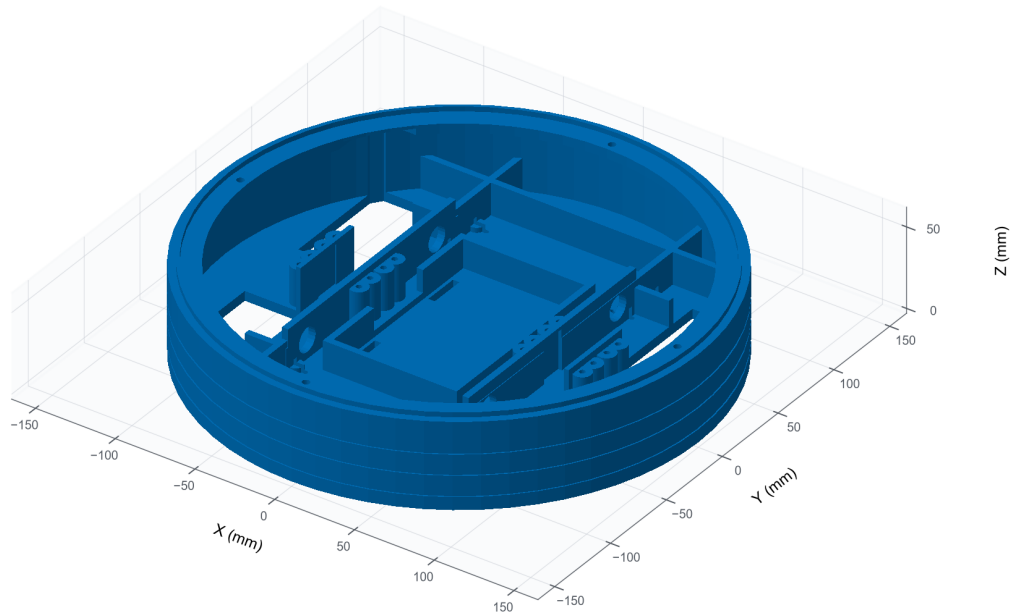
For service, isolate power, unplug labelled harnesses and remove the required horizontal joint screws. Lift whole body sections vertically. Remove the neck and disconnect the four arm connectors to service the shoulder; fit the internal mechanisms while its top is open. The motor plates can be removed separately for base service. Back off the head jackscrew before lifting the dome. Remove its crown nut with the specified socket and lift by the dome rim, never by the eye or lamps. Refit socket liners and repeat the motion checks after service.

Regenerate with `C:/Python314/python.exe scripts/export_cad.py --jobs 3`, then check with `C:/Python314/python.exe scripts/export_cad.py --check-only`. The exporter uses a 600-second timeout for the detailed combined skirt and 240 seconds for every other part. Its assembly views import the released meshes. Re-run the actual H2D slice checks whenever a mesh changes.

Single-piece circular motor base

Dalek ROUND-9 / base

Actual STL geometry | Purchased hardware uses nominal envelopes



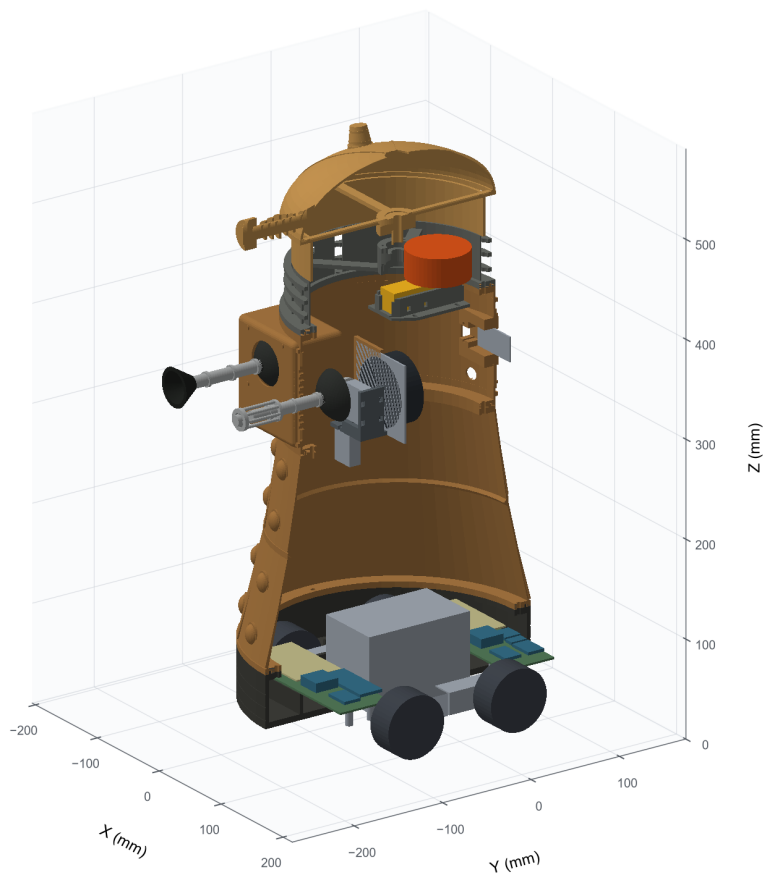
Actual ROUND-9 STL geometry | Dimensions in mm | +Y is rear; -Y is front

The floor, circular perimeter, ribs, motor pockets and battery support form one connected print. The base is 300 mm in diameter before its brim.

Internal assembly view

Dalek ROUND-9 / section

Right shell half omitted to expose internal components



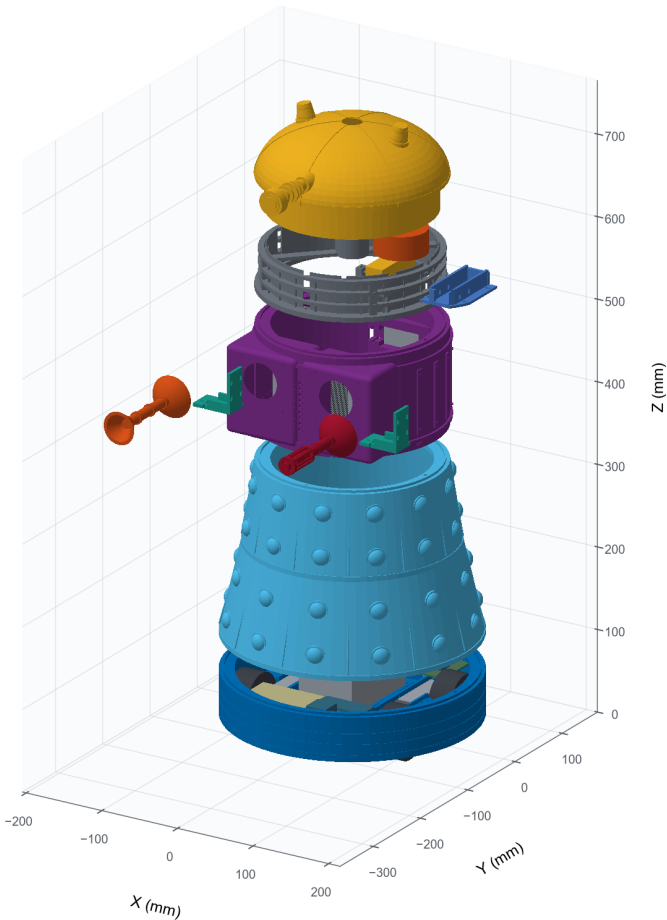
Actual ROUND-9 STL geometry | Dimensions in mm | +Y is rear; -Y is front

Section through the native CAD. The continuous base supports the motor pockets, battery and vertically stacked body.

Vertical stacking order

Dalek ROUND-9 / exploded

Actual STL geometry | Purchased hardware uses nominal envelopes



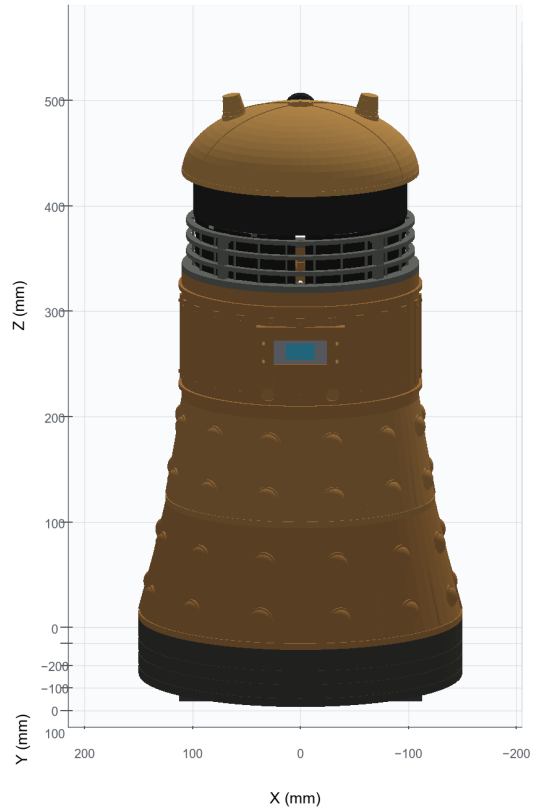
Actual ROUND-9 STL geometry | Dimensions in mm | +Y is rear; -Y is front

Each body tier is one complete print. Registers align the stack; accessible fasteners retain the joints. The gaps in this view are intentional.

Rear display and access

Dalek ROUND-9 / rear

Rear-mounted original TTGO T-Display



Actual ROUND-9 STL geometry | Dimensions in mm | +Y is rear; -Y is front

The TTGO display faces outward. Keep the USB opening, operator switches and service fasteners accessible.

Firmware and Wi-Fi controller

This firmware targets the original 1.14-inch LilyGO TTGO T-Display with a classic ESP32, 135 × 240 ST7789 display and 4 MB flash. It does not target the T-Display S3 or the AMOLED version. The display faces outward through the robot's rear window. Landscape rotation is 1; use rotation 3 in `src/main.cpp` if the finished mount is reversed.

The software builds for the specified hardware. Its physical operation must be commissioned on the assembled robot. A successful compiler check does not establish stopping distance, servo centre, current draw, Wi-Fi range, speaker level or battery measurement accuracy.

Build and load

1. Install PlatformIO Core or the PlatformIO extension for Visual Studio Code. The pinned packages in `firmware/platformio.ini` install on the first build. Internet access is required for that build only.
2. Turn the physical actuator stop switch OFF. Disconnect the battery before connecting USB. Do not connect two 5 V sources to the board together unless the circuit diagram's power isolation is in place.
3. Use a data-capable USB cable. From the project directory run:

```
Set-Location C:\dev\robots\dalek\firmware

pio run

pio run -t buildfs

pio device list
```

4. Read the ESP32 serial port from the last command. Replace COM7 below with that port:

```
pio run -t upload --upload-port COM7

pio run -t uploadfs --upload-port COM7
```

5. Reset the board. The display must show DISARMED. If automatic flashing does not start, hold the board's BOOT button, start upload and release BOOT after the connection begins. The other button, GPIO35, is the stop/setup button.
6. Disconnect USB before restoring the normal battery supply. Keep the wheels off the floor until all commissioning checks below pass.

There is one factory application partition, 0x10000-0x1ffff (2,031,616 bytes). LittleFS occupies 0x200000-0x3ffff (2,097,152 bytes). Uploading LittleFS replaces the web assets and MP3 files in that partition. Network credentials are in a separate NVS partition. There is no wireless firmware update, cloud service, internet dependency or scheduled background job.

Pinned dependencies: Espressif32 PlatformIO platform 6.5.0, Arduino-ESP32 2.0.14, TFT_eSPI 2.5.43, Adafruit PWM Servo Driver Library 3.0.2, Adafruit BusIO 1.16.1 and ESP8266Audio 1.9.7. The audio dependency is GPL-3.0; retain its notices and supply corresponding source with redistributed firmware binaries. Later ESP8266Audio releases use different ESP32 I2S APIs, so update the platform and audio library together only after testing.

First connection

The board makes a WPA2 access point named DALEK-xxxx. It generates a device-specific 12-character key on first boot and saves it locally. The rear display shows the key. Firmware does not print keys or station passwords to the serial terminal. Join this access point using the displayed key, then open `http://192.168.4.1`. Enter the same key in the control page.

To join a router, open Wi-Fi network settings on the page. Enter the exact network name and its 8-63-byte WPA2 password. This classic ESP32 uses 2.4 GHz Wi-Fi. The password is sent in an authenticated POST request and saved in local NVS. Read the LAN address on the rear display and browse to that address from the same local network. No credentials are present in the repository.

The setup AP remains available while the robot connects to a router. This also provides fallback if the router is missing or its credentials change. Moving between AP and router connections, or losing either connection, disarms motion. Reconnecting never arms the robot. The AP supports two clients, but only one control lease can be active. Keep one control browser open.

Press the rear GPIO35 button to disarm. Hold it for five seconds to erase only the saved router credentials and return to AP operation. The device access key stays the same. The rear key also authorizes robot control for users on the router network. Do not share it widely or expose port 80 to the internet. The local interface uses HTTP; Wi-Fi encryption protects the wireless link, but this is not an end-to-end encrypted service.

Controls

Press Arm controls, then hold a movement button. Forward uses PWM 110/255, reverse 90/255 and turns use 50/255 on the inside wheel pair and 120/255 on the outside pair. Release disarms. Arm again for the next movement. Tight pivots are omitted from the supplied interface because four rubber wheels can scrub heavily against the floor. The requested TT motors have limited torque. Use a smooth, level indoor floor and gentle arcs.

The arm checkbox enables two-axis circles. Radius is 0-8 degrees and frequency is 0.10-0.80 Hz. The default is 8 degrees at 0.40 Hz. Eight degrees is the concealed gimbal's clearance limit; the browser and API reject a larger radius. One side runs half a cycle behind the other. The head slider commands the TT friction motor's drive level and direction; zero removes PWM. Displayed 100% maps to the conservative 100/255 PWM ceiling. It is not a measured speed. Hold Arms / head only to animate without ground-wheel movement. These settings also apply while a drive button is held. Sliders alone do not cause movement.

Speak plays a selected MP3 locally. Silence stops the decoder. Audio operates while the robot is disarmed. All audio stays in the fixed body, so the head can turn indefinitely without twisting an electrical cable. There is no head position sensor or automatic front-facing home position.

The browser must stay visible and focused. Releasing a pointer, losing pointer capture, cancelling a touch, changing tabs, locking the screen, closing the page or pressing Escape requests a stop. Loss of a command response also disarms the interface. Browsers can delay or discard stop packets; the board's independent timeout covers that case.

Electrical contract

Read `electronics-research.md` (`electronics-research.md`) and the power circuit (`../electronics/01-power.svg`), controller, drive and audio wiring (`../electronics/02-controller-drive-audio.svg`), servo wiring (`../electronics/03-servos.svg`) and circuit details (`../electronics/04-circuit-details.svg`) before wiring. All ESP32 signal pins are 3.3 V. Servo signals pass through the specified 74AHCT125 buffers to reach 5 V logic. Never feed 5 V into an ESP32 input.

Function | ESP32 GPIO | Connection

I2C data / clock | 21 / 22 | PCA9685 SDA / SCL, logic VCC 3.3 V

Servo disable | 27 | PCA9685 OE and U8 arm-buffer active-low output enables; U9 uses PCA4/5

Left forward / reverse | 25 / 26 | Left DRV8833 AIN1+BIN1 / AIN2+BIN2

Right forward / reverse | 32 / 33 | Right DRV8833 AIN1+BIN1 / AIN2+BIN2

Motor enable | 12 | All three DRV8833 SLP pins; 10 kΩ pull-down

Head PWM | 2 | 20 kHz to U9 inputs2/5;10 kΩ pull-down

Audio bit clock / word clock / data | 17 / 13 / 15 | MAX98357A BCLK / LRC / DIN

Actuator power sense | 36 | 5 V actuator rail through 10 kΩ top / 15 kΩ bottom divider

Battery voltage sense | 39 | Battery through 100 kΩ top / 22 kΩ bottom divider, 100 nF from ADC to ground

Stop / forget router | 35 | Existing rear board button

PCA9685 address is 0x40. Servo channels 0-3 are left yaw, left pitch, right yaw and right pitch. Channels 4/5 are static active-low head gate enables, not servo connectors. U8 buffers the four servo signals from 5V_SERVO. U9, powered from 5V_MOTOR, routes GPIO2 PWM to the selected U11 bridge input. PCA4/5 HIGH disables the corresponding U9 gate; LOW enables it. Each gate has a 10k pull-up to 3V3. U9 outputs reach U11 AIN1/AIN2 through 220ohm resistors, with 10k pull-downs at the driver. See circuit sheet03 for physical DIP pins. The PCA stays at 50Hz; the motor receives 20kHz hardware PWM from the ESP32.

U5/U6 each run two ground motors, with input pairs joined. U11 bridgeA runs the head; tie BIN1/BIN2 low and leave BOUT1/BOUT2 open. Every used bridge retains its stock 1A current limit. Never parallel bridge outputs.

Startup writes and verifies PCA MODE2=0x06: OUTDRV=1 and OUTNE=10. Global OE HIGH then makes PCA outputs high-impedance. R7/R8 pull both head-gate enables HIGH. The default OUTNE=00 would instead enable both active-low gates. A failed write or readback prevents arming. Keep this setting when changing PCA libraries.

GPIO27 needs a 10k pull-up to 3.3V. Remove the PCA breakout's onboard OE pull-down as specified in the electronics guide. GPIO12 and GPIO2 each need 10k pull-downs because they are boot strap pins and must default to disabled motion. GPIO15 connects only to the amplifier DIN input without external bias. GPIO36/39 are input-only and use the specified dividers.

The firmware reserves GPIO5/16/18/19/23/4 for the integrated display. GPIO0 is the existing BOOT button. GPIO34/14 remain reserved for the board's original battery measurement circuitry. The robot's external 12 V pack does not connect to the board's single-cell battery socket.

Stop behavior and limits

The motion task runs independently of the HTTP server and MP3 decoder, with a nominal 10 ms period. It is watched by the ESP32 two-second task watchdog. External enable resistors keep the actuator signals disabled during reset. This is a hobby control system, not a certified safety controller. The physical stop switch cuts actuator power without relying on Wi-Fi or software.

A command lease is granted only after explicit arming. The robot requires a complete valid command at least every 500 ms. The supplied browser sends a fresh command snapshot every 100 ms and aborts a command request after 300 ms. Each arm uses a new random lease; every command needs a strictly increasing sequence number. Old leases and duplicate commands do not refresh the timeout. An expired lease is checked before accepting another command, so a late request cannot revive movement. Invalid control parameters disarm immediately. Software drive values are limited to $\pm 150/255$.

Ordinary wheel speed changes ramp by at most five PWM counts per nominal 10 ms. A requested reversal first ramps to zero, waits 100 ms, then ramps the other way. A stop, fault or disarm immediately drives SLP low and both inputs low; it does not wait for the ramp. This lets the wheels coast. It does not provide a mechanical brake, and stopping distance must be measured with the final mass.

The head motor uses the same five-count ramp step with a 100/255 ceiling. Direction reversal ramps to zero, coasts for 100ms, then ramps up. A brief zero command or quick disarm/re-arm cannot bypass that interval. A zero head command immediately removes head PWM. Before changing the selected U9 gate, firmware sets GPIO2 PWM to zero, disables both gates using checked I2C writes, enables one gate and then restores bounded PWM. The shared SLP line is held steady during operation; PWM never drives SLP.

The PCA receives four checked servo writes every 20ms and checked static-gate writes when head direction changes. Any failed write latches an I2C fault, sets head PWM zero, sets all driver SLP pins low and disables servo OE. Missing PCA at boot prevents arming. No automatic motion recovery follows power, I2C, Wi-Fi or battery faults. Head PWM stops immediately on software disarm; the old head-servo neutral pulse is removed. All motors coast after electrical disable. Physical stopping distance and head coast angle still need measurement.

The specified battery is the Bioenno BLF-1206A 12 V nominal 6 Ah LiFePO4 pack. The software multiplies the ADC voltage by $(100 + 22) / 22 = 5.54545$ to recover pack voltage. It smooths readings at 10 Hz. Below 11.2 V for one second, it disarms; voltage must recover to at least 12.0 V before re-arming is allowed. A disconnected ADC or an implausible voltage also prevents arming. These thresholds are conservative operating limits, not a battery state-of-charge meter, a charger or a replacement for the pack's protection circuit. Calibrate the ADC against a multimeter before use.

Servo and sensor calibration

All calibration values are in `firmware/include/config.h`. Keep the wheels lifted and the servo horns removed for initial checks.

1. Check 5V rails with actuator power disabled. During ESP32 reset verify all three SLP pins and GPIO2 are low, and PCA OE is high.
2. Measure battery voltage with a multimeter. Read the displayed voltage. Set `PACK_ADC_CORRECTION` to measured voltage divided by displayed voltage, rebuild, and recheck at two battery voltages. Do not lower a cutoff to hide a wiring error.
3. Leave arm and head commands at zero. Enable actuator power, arm, and hold Arms / head only. The four positional servos must settle near their centres. Fit each horn with its joint centred and no mechanical load, then stop before fitting links. The initial centres are 1500 μ s; change `ARM_CENTER_US` in small steps only if needed.
4. Keep the head tyre clear of its track and command zero. GPIO2 and both U11 bridge inputs must remain low. Before attaching the motor, scope AIN1/AIN2 at a low head setting: only one input may carry 20kHz PWM. Check both directions and a reversal interval of at least 100ms with no PWM. There is no neutral pulse or neutral trimmer.
5. Start arm circles at radius 2 degrees and 0.10 Hz. Confirm that yaw and pitch are both active. Confirm that neither joint touches a stop. Increase to 8 degrees and 0.40 Hz. The pulse envelope stays within 1200-1800 μ s, but mechanical clearance is the limiting factor. Reduce the maximum radius if the actual links require it.
6. Turn power off. Set the head mount to light tyre contact, then test in each direction. Increase pressure only enough to turn the freely rotating head. Preserve slip if obstructed. `HEAD_DIRECTION` reverses the sign and `dalek: :HEAD_LIMIT` is 100/255. Do not raise this ceiling to overcome a jam. The slider is a drive-level command, not an angle or measured speed.
7. Lift all wheels. A forward command must rotate all four wheels toward forward travel. If one motor is reversed, swap only that motor's two output wires. `LEFT_DIRECTION` and `RIGHT_DIRECTION` reverse an entire side if necessary. Stop and remove actuator power before rewiring.

8. Play a phrase. The fixed digital gain is 0.28, below the design ceiling of 0.35. MAX98357A hardware gain must be 3 dB as specified in the circuit guide. Do not raise gain above 0.35 with the 8 Ω 1 W speaker and specified logic supply. Digital gain is not a calibrated wattmeter; verify clean sound, supply stability and acceptable speaker temperature.

Acceptance checks on the completed robot

Perform these with a helper at the physical stop switch, first with wheels lifted, then at low speed on a clear floor. Record the results in the assembly commissioning sheet.

- Power up, reset and reconnect USB: no wheel or servo motion before an explicit arm command.
- AP mode: router absent, page and every local voice phrase work through 192.168.4.1.
- Station mode: join the private router, read the rear LAN IP, load the page and drive through that address.
- Release the movement button: wheel outputs disable and the page becomes disarmed. Measure actual coasting distance with the final shell and battery fitted.
- While moving, disable the phone's Wi-Fi: commanded drive must cease within the 500 ms lease timeout plus control-task scheduling delay. The robot must remain disarmed after reconnection.
- While moving, lock the phone or change tabs: motion stops. Unlocking does not resume the prior command.
- Press physical stop: motor and servo supply must lose power even if the ESP32 keeps running. Releasing the switch does not restart movement.
- Press the rear GPIO35 button while holding a browser control: motion stops and does not restart without a new arm.
- With actuator power off, disconnect the PCA9685, then boot: I2C fault appears and Arm is refused. Reconnect only with power removed.
- Use an adjustable current-limited bench supply in place of the pack for the voltage check. At 11.0 V for more than one second the software must disarm. At 11.5 V it must refuse a new arm. Above 12.0 V it may accept a new explicit arm.
- Run arm circles and head motion while playing all phrases; no reset, supply sag, decoder stall or joint contact is acceptable.
- Complete gentle forward arcs on the intended floor. Measure individual motor current, case temperature and stopping distance. Do not accept a stalled inside wheel or prolonged current-limit operation.

Automated checks

From firmware/:

```
pio run
pio run -t buildfs
node --check data/app.js
node test/test_ui.js

$env:PATH = 'C:\msys64\mingw64\bin;' + $env:PATH
g++ -std=c++11 -Wall -Wextra -Werror -I include test/test_control.cpp -o $env:TEMP\dalek-control-test.exe
& $env:TEMP\dalek-control-test.exe
```

The host C++ test runs the same control-state code used by the ESP32. It covers strict parsing, boot lockout, expired/replayed leases, reconnects, unhealthy-input disarm and millisecond rollover. It accepts an 8-degree arm radius and rejects 9 degrees. Head regressions check the PWM ceiling, immediate zero/fault stop, and reversal delay through zero, re-arm and clock rollover. The Node test exercises the actual browser script with a simulated DOM and transport. Hardware routing, timing and stopping still need physical commissioning.

Sources

- [LilyGO original T-Display board, schematic links and pin allocation.](#)
- [PlatformIO LilyGo T-Display board definition.](#)
- [TFT_eSPI original T-Display setup.](#)
- [Adafruit PCA9685 driver source and checked setPWM result.](#)
- [ESP8266Audio version 1.9.7 source and license.](#)
- [ESP8266Audio current API compatibility notes.](#)

Dalek electronics design and source record

Original research: 2026-09-11. Head friction-drive revision researched 2026-09-12 against Adafruit, LILYGO, TI and NXP primary sources. This is a build design, not a claim that a robot was physically assembled or tested. Listed prices are USD before shipping and tax. Stock is a page observation, not an order reservation. The bill of materials separates checked prices from allowances.

Selected build

The controller is the original 1.14-inch TTGO T-Display ESP32. It is mounted on the rear with its screen visible. Four Adafruit 3777 motors each drive one Adafruit 3766 ground wheel. A fifth identical motor and wheel turn the head by adjustable friction contact. Each motor has a separate current-limited H-bridge. Two positional servos inside the shoulder move each arm in yaw and pitch. Their phased movement draws a circle with the arm tip. No servo is mounted on the exposed arm. The head motor stays fixed to the body. No wires enter the rotating head.

Power comes from one removable Bioenno BLF-1206A 12 V 6 Ah LiFePO4 pack. Three independent converters produce 5 V for drive motors, 5 V for servos, and 5 V for controller/audio. The physical ACTUATORS OFF switch removes input power from both motion converters. Logic and the display remain on. All circuit grounds join at a power distribution star.

The authoritative connection set is `electronics/wiring.csv` together with the circuit SVGs. Use the component references from those files. Pin numbers on the TTGO are GPIO numbers. Pin numbers on U8/U9 are DIP package pin numbers.

Required motors and wheels

Adafruit 3777 accepts 3-6 V. The vendor measured 160 mA unloaded and 1.5 A stalled at 6 V on a sample. At 4.5 V it measured 155 mA unloaded and 1.2 A stalled. The listing gives 0.8 kg·cm stall torque at 6 V, a 48:1 gearbox, 70×22×18 mm body and 30.6 g mass. There is no encoder. All five motors use 5 V. Five motors cost \$14.75 at the checked \$2.95 each. These values do not establish continuous load torque. A 5 V stall-current value was not published. [Adafruit 3777](#)

The required orange/clear 3766 is a press-fit TT wheel, 63 mm diameter by 29 mm wide, 38 g. The checked price is \$1.50 each. Adafruit marked it out of stock. This design retains the user's exact wheel. Obtain the wheels before accepting the final motor-to-shell fit. Do not buy a different wheel from a similar photograph. [Adafruit 3766](#)

A nominally stronger DRV8871 was rejected because its minimum motor supply is 6.5 V. That is above the required motor's 6 V maximum. Feeding the motors an overvoltage and merely lowering PWM duty is not used here. [Adafruit DRV8871](#)

Three DRV8833 boards support the 5 V motor supply. U5/U6 each run two ground motors. U11 bridge A runs the head; tie U11 BIN1/BIN2 to ground and leave its B outputs open. Each stock 0.2 ohm sense resistor sets approximately 1 A limit through 0.2 V/R. Keep the factory resistors and bypass jumpers open. On U5/U6 only, tie AIN1 to BIN1 and AIN2 to BIN2. Never join bridge outputs. All three SLP inputs share GPIO12 and its external 10k pull-down. A low SLP disables all five motors. The current limit does not make an indefinite stall acceptable. [DRV8833 product](#), [DRV8833 pinouts](#)

The four-wheel skid-steer layout has no precise speed or distance measurement. The original 3 kg planning target was not a motor manufacturer's payload rating. Use the current print report and the complete hardware mass, then weigh the finished robot. A slicer result does not prove loaded turning ability. Use hard flat indoor flooring, broad moving arcs and the physical current, temperature and stop tests below. Wide silicone tyres can resist turning. Do not increase voltage or remove current protection to force a stalled build to move.

Servos and movement

The four arm servos are MG92B units, Adafruit 2307. The listing specifies 5 V signals as well as 5 V power. Direct 3.3 V PWM is therefore not the chosen wiring. The listed torque is 3.1 kg·cm at 4.8 V and 3.5 kg·cm at 6 V; weight 14 g; envelope 36×12×31 mm; 20-tooth horn spline. The listing also says this servo has no mechanical end stops. Use its supplied horn rather than printing a nominal spline. Begin with a small software travel window and check every linkage. [MG92B](#)

The head uses a fifth TT3777 and 63 mm wheel3766. Its tyre presses against the circular head track. The screw-adjusted mount sets contact pressure with power removed. Bearings carry the head mass. The motor mount and wire remain stationary. Begin with light contact, then increase only until the head turns reliably. The contact must still slip when obstructed; added pressure is not a substitute for correcting a tight bearing or track. There is no encoder, so the slider commands power and direction rather than an angle or measured speed. [TT3777](#), [wheel3766](#)

U10 is Adafruit 815 PCA9685. Use 3.3 V VCC and its default I2C address 0x40. The board's I2C pull-ups then stay at the ESP32 logic voltage. Leave the V+ rail disconnected because servo power is distributed separately. The board's output-enable pin is active low and normally pulled down. Remove that pull-down resistor, then fit an external 10k resistor from OE to 3.3 V. Identify it by continuity to OE and GND against the actual PCB revision; do not remove a resistor by guessed reference number. Measure the result before connecting servos. [PCA9685 pinouts](#)

U8/U9 are 74AHCT125 buffers. Power U8 from 5V_SERVO and U9 from 5V_MOTOR. Their TTL-compatible inputs recognize 3.3 V; outputs follow their own 5 V rail. U8 makes the four arm signals. U9 routes head PWM to the selected bridge input. Use AHCT, not HC, and check the actual chip marking. TI specifies input leakage with VCC=0 through 5.5 V; this avoids powering a disconnected actuator rail through controller signals. C8/C9 are 100 nF at the respective chip supplies. [Adafruit1787](#), [TI SN74AHCT125](#)

Servo channel order is 0 left yaw, 1 left pitch, 2 right yaw and 3 right pitch at 50 Hz. With horns removed, start at 1500 μ s. Fit arm horns at the mechanical centre. Begin circles at 2 degrees and 0.10 Hz. The default and maximum radius are 8 degrees after hidden-linkage clearance checks. The normal frequency is 0.40 Hz. The old 12-degree option is removed because it reduces case clearance in this concealed layout. Pitch is 90 degrees out of phase with yaw. The arm tip draws a cone; the shaft does not rotate continuously.

The original board exposes only one unused output pin, GPIO2, after the existing display, audio and controls are allocated. It feeds 20kHz PWM to U9 inputs2/5 with a 10k pull-down. PCA channels4/5 drive U9 active-low enables1/4 as static full-on/full-off levels. HIGH disables a gate; LOW passes PWM. R7/R8 pull the enables to 3V3. U9 outputs3/6 reach U11 AIN1/AIN2 through 220ohm resistors with 10k pull-downs. Spare U9 inputs9/12 are grounded; enables10/13 use 5V_MOTOR. Firmware writes and reads back MODE2=0x06 (OUTDRV=1, OUTNE=10). Therefore global OE HIGH makes PCA outputs high-impedance and both head gates disable through their pull-ups. The factory OUTNE=00 is unsuitable because it drives these active-low gate enables LOW. No 50Hz PWM drives the motor; GPIO2 supplies 20kHz independently. [LILYGO pin map](#), [NXP PCA9685 MODE2 and full-on/full-off controls](#)

Firmware limits head duty to 100/255. Before a direction update it removes GPIO2 PWM, writes both gates disabled and checks every I2C acknowledgement, then enables the selected gate. Reversal ramps to zero and waits 100ms before applying opposite polarity. A zero command, disarm or fault removes PWM immediately. Timeout, rear STOP, Wi-Fi loss, low battery and I2C failure disable the shared SLP line. The 100ms interval also applies after a brief zero command or quick re-arm. The driver uses fast-decay PWM, with one input pulsed and the other low. SLP is held steady during operation; it is not used as a PWM input. [TI DRV8833 control table](#)

Power and battery

The selected BLF-1206A provides 12 V nominal, 6 Ah, 72 Wh, 12 A continuous discharge, and internal protection and cell balancing. Published mass is 0.7 kg. Its metric and inch dimensions differ: 108×64×69 mm versus 4.4×2.6×2.8 in, or 111.8×66.0×71.1 mm. Use the larger values for fit. The round base rotates the pack 90 degrees: its padded design envelope is 70 X × 114 Y × 76 Z mm, inside a 76 X × 116 Y mm pocket. Route the factory connector up the left notch, away from motor ends. Keep the PP30 discharge connector intact. [Bioenno BLF-1206A](#)

Use only the matched BPC-1502DC charger for this build: 14.6 V, 2 A with the specified DC charge plug. Switch off, unplug the PP30 discharge connection, and remove the pack from the printed shell before charging. Charge on a nonflammable surface with the pack and charger visible. Do not charge through TTGO's lithium battery connector. This robot has no onboard charging circuit. [Bioenno matched charger](#)

U2/U3 are Pololu 4091 D36V50F5 regulators. Their fixed 5 V outputs, input range and current capability give substantially more supply margin than the rejected low-voltage pack arrangement. The manufacturer's output-current rating depends on input voltage and temperature. The 5.5 A name is not an unconditional guarantee inside a printed enclosure. Leave the boards ventilated and solder thick power conductors directly to their duplicated power pads. Do not send more than 3 A through one ordinary header pin. The board is 25.4×25.4×9.5 mm and has M2 mounting holes. Leave EN, PG and VRP unconnected. [Pololu 4091](#)

U4 is the Adafruit 4739 MPM3610 fixed 5 V regulator. It is rated for 1.2 A and up to 21 V input. The checked price is \$6.95. This lower-power converter is sufficient only with the specified audio limit. CAD from the manufacturer's board repository gives 10.16×17.145 mm outline and a 2.5 mm mount hole at (5.08, 14.605) mm. The 5 V variant uses the same outline; verify the actual board before fastening. [MPM3610 product](#), [Adafruit board CAD](#)

Use NKK S1A switches for MAIN and ACTUATORS OFF. Their source rating is 20 A at 30 VDC; the panel opening is 12.5 mm. ON joins terminals 1 and 3. The small Adafruit covered toggle was rejected because Adafruit recommends only 1-2 A despite the 20 A marking. Each battery/branch fuse has a stated value. The main 7.5 A fuse protects the harness close to the battery. Branch fuses are 3 A at each motion converter input and 1 A at the logic converter input. These are wire-fault protection, not a precision current limiter or motor thermal model. [NKK S1A](#), [Adafruit covered toggle](#)

Current allowances at 5 V are 5 A motors (four wheel bridges plus one head bridge), 4 A for four servos, and 0.9 A logic/audio. The two discharge resistors add 0.5 W, giving 50 W total and 5.25 A estimated pack current at 11.2 V and 85% conversion efficiency. The motor regulator allowance including its bleeder is 5.05 A; the servo regulator allowance is 4.05 A. These are allowances, not measurements or unconditional regulator ratings. The 4 A servo budget uses 1 A per unit because authoritative stall-current data is missing. Measure actual units with head, arms, drive and audio active. Keep the existing 7.5 A main fuse and 3 A / 3 A / 1 A input branch fuses. Stop if a rail falls below 4.8 V, a fuse opens or a driver shuts down. Never increase fuse or sense-current limits to force movement.

The 100 ohm 1 W discharge resistors each consume 0.25 W. They reduce residual actuator voltage after S2 opens. Capacitors still contain energy for a fraction of a second. The disconnect causes coasting; it is not an instantaneous mechanical brake or a certified emergency-stop system. Prove that both actuator rails fall below 0.5 V within 1 second during the commissioning test, including the no-load case.

Controller and Wi-Fi connections

The original board uses ST7789 display pins 19, 18, 5, 16, 23 and 4. Buttons use 0 and 35. Its onboard single-cell battery measurement is GPIO34 with GPIO14 power control; that circuit must not measure the 12 V pack. The chosen separate ADC is GPIO39(SVN). GPIO36(SVP) reads actuator power state. Both are input-only. The pin map source confirms those pins are exposed. [LILYGO pin map and source](#), [manufacturer pin-map image](#)

External GPIO assignments are I2C 21/22, servo-disable 27, motor-enable 12, left drive 25/26, right drive 32/33, head PWM 2, audio 17/13/15, battery 39 and actuator sense 36. GPIO12 and GPIO2 each have a 10k pull-down to preserve their low boot strap state. GPIO15 connects only to the amplifier input without added bias. No GPIO receives 5 V. Use 2.4 GHz Wi-Fi. The firmware supplies an AP and can join an existing network. Keep the rear antenna away from the battery, motor cans, metallic paint and fasteners.

The pack divider is 100k top/22k bottom, 1% resistors, with 100 nF at GPIO39. At 14.6 V it gives 2.633 V. Multiplier is 122/22. A 100k/33k divider is NOT used; it would exceed the intended ADC range on this pack. A second 10k/15k divider maps 5V_MOTOR to 3.0 V at GPIO36, with 100 nF filtering. Firmware stops on a sustained 11.2 V reading and permits re-arm only above 12.0 V. LiFePO4 voltage does not provide a reliable linear charge percentage. Display voltage or a conservative low-battery state, not a claimed precise fuel gauge.

Sound

The ESP32 decodes the supplied MP3 files and sends I2S to Adafruit 3006 MAX98357A. It needs BCLK on GPIO17, LRC on GPIO13 and DIN on GPIO15. It does not require MCLK. OUT+ and OUT- go only to the speaker. Neither speaker lead connects to ground. Use a 100k resistor from GAIN to 5V_LOGIC for 3 dB gain, and leave SD at its factory mono-mix configuration. [MAX98357 pinouts](#)

The speaker is Adafruit 1313, 8 ohm 1 W, 77.8 × 77.8 × 25.49 mm. Its four mounting tabs are 60 mm apart. It costs \$1.95 and was in stock. The supplied firmware uses fixed digital gain 0.28. This is a design setting, not an acoustic measurement. Verify that continuous test audio is below 2.83 Vrms across the 8 ohm load, using a suitable differential measurement and audio-band filtering. Never attach a grounded oscilloscope clip to an amplifier output. Normal spoken clips have lower average power than a continuous test tone. [Adafruit 1313](#)

Board placement on the round base

Use two insulating FR4 plates, each 160×56×2 mm, centred at (X0,Y-93) and (X0,Y+93). Front plate bounds are X-80..80 and Y-121..-65. Rear bounds are X-80..80 and Y65..121. Coordinates are measured from the robot centre; +Y is rear. Base mounting points are X±54 at Y-93,-69,+69,+93. Drill PCB holes in the FR4 using the actual boards as templates. Do not assume a mounting-hole pitch from a bounding box.

The front plate holds the buffer prototype board at (-35,-93), PCA9685 at (40,-80), U5 left driver at (65,-106), and U2 motor regulator at (26,-107). The rear plate holds the divider prototype at (-35,93), U3 servo regulator at (24,81), U6 right driver at (62,78), U11 head driver at (62,104), U7 amplifier at (23,107), and U4 logic regulator at (41,105). These are board-centre coordinates in millimetres. Keep the board long side along X except the logic regulator, whose 17.145 mm side follows Y.

The two prototype outlines are 81×51 mm; PCA is 62.5×25.4 mm; each driver is 26×18 mm; each high-current regulator is 25.4×25.4 mm; the amplifier is 19.4×17.8 mm; logic regulator is 10.16×17.145 mm. The rectangular envelopes fit without overlap. Connector bodies, solder joints, nuts and wiring still require physical clearance checks. Solder high-current leads directly to the regulator pads. Route wires above the boards with strain relief and keep heat-producing parts ventilated. No drilling may damage a trace. [Adafruit PCA9685](#), [DRV8833](#), [MAX98357A](#), [prototype board](#), [Pololu regulator](#)

Harness assembly

1. Disconnect the battery. Fit the switches, fuse holders and ground-star block. Label every rail and both switch OFF positions. Protect all exposed lugs with heat shrink.
2. Fit U2/U3 on insulated standoffs with air space around them. Fit U4 and the two prototype boards. Fit U1 in the rear holder with the display facing outward. Keep the USB connector accessible.

3. Assemble the buffer board without ICs installed. Use DIP14 sockets. Check U8 pin14 to5V_SERVO, U9 pin14 to5V_MOTOR, and both pin7 toGND. Check every channel against sheet03. Add one100nF capacitor per socket. Fit signal pull-downs, gate pull-ups and220ohm output resistors.
4. Modify PCA OE as described above. With the board unpowered and isolated, check that OE is not still tied to ground through its original resistor. Add the external 10k to 3.3 V. Record the actual board revision and removed resistor location for later service.
5. Wire the power branches in 18 AWG. Wire one 22 AWG positive/ground pair from the servo distribution block to each servo plug. Do not route aggregate servo current through the PCA header rail or prototype-board strips.
6. Wire each motor to its own bridge. Add 100 nF directly at each motor's solder tabs. Retain only short lengths of the supplied 28 AWG leads. Twist the two motor conductors. Add strain relief near the plastic gearbox, not on the motor tabs.
7. Wire signals from the CSV. Keep I2C and I2S away from motor wiring. Add service loops between vertically stacked sections and the rear board mount. Keep wires clear of wheels, the head friction track and hidden arm links. The rotating head has no wiring. U9 and the head motor both use5V_MOTOR; U8 and the four servos use5V_SERVO.
8. Fit and label the removable J_USB_ISO run-power connection. To program: switch MAIN off, unplug the battery, open the run-power link, then attach USB. Unplug USB before reconnecting the link and battery. No external battery is connected to the TTGO JST socket.

Bench acceptance before the shell is closed

Use a current-limited bench supply for first power where available. Otherwise leave motors and servos unplugged until polarity and rail checks pass. Raise all four ground wheels. Keep hands clear of hidden links and the head friction contact.

1. With no power, check for shorts between each positive rail and GND. Check for no direct connection between the three 5 V outputs. Check capacitor polarity and fuse values. Check each power connection by meter, not wire color alone.
2. MAIN on and ACTUATORS off: measure logic 5 V; verify rear screen boots. Verify both actuator rails discharge below 0.5 V within 1 second. Check no servo power appears through a signal wire.
3. ACTUATORS on but disarmed: measure all three rails 4.8-5.2 V. Confirm the battery reading agrees with a meter within 0.2 V after calibration. Verify GPIO36 is about 3.0 V.
4. Hold TTGO reset. Verify all three SLP pins and GPIO2 below0.3V, and PCA OE above3.0V. Motors must not run. Release reset; the robot stays disarmed. Repeat power cycles with ACTUATORS on.
5. Fit one motor at a time. Verify direction and stop at low duty, with ground wheels raised and head tyre clear of its track. Before attaching the head motor, scope U11 AIN1/AIN2 at low command. Only one input may have20kHz PWM. Maximum duty must be100/255 and a direction reversal must include at least100ms with both inputs low. Do not deliberately stall a motor.
6. Fit four arm servos with horns removed. Check centres and mapping. Attach centred horns, then make slow small circles while the shoulder is open. Fit the head track after checking bearings rotate freely. With power off, adjust tyre pressure only enough for reliable drive. Zero head command must remove PWM; there is no servo-neutral adjustment.
7. At low speed, close the browser, turn off Wi-Fi and disconnect the control device in separate tests. The500ms timeout must remove wheel and head drive and disable arm signals. Reconnect requires explicit arming. Test

physical ACTUATORS OFF while driving the head and ground wheels; both motion supplies must discharge while the display remains on. Measure actual head coast angle.

8. Simulate low battery with an adjustable supply rather than over-discharging the pack. Confirm sustained 11.2 V causes stop, 12.0 V is required before re-arm, and an invalid ADC prevents arming.
9. Weigh the finished robot. First drive on a hard flat floor at low duty. Test straight motion and broad forward turns. The supplied control interface intentionally uses arcs rather than in-place turns. Log loaded pack voltage, each 5 V rail minimum and motor-driver temperature. No rail may drop below 4.8 V; there must be no reset or repeated thermal shutdown. Stop if the robot binds while turning. A smaller/lighter shell or lower-friction floor is required if the specified motors cannot turn it reliably.
10. Run 15 minutes of representative driving and animation with pauses. Confirm regulator and driver package cases remain below 60°C with a thermometer. Motors, servos and printed mounts must stay below the assembly guide's more conservative 50°C limit. Wire joints must remain cool and wheels must not rub. These are project acceptance limits, not claimed vendor maximums. Test audio with drive and arms active. Check there is no reboot or audible power noise.

Calculations and limits

electronics/calculations.json is generated from the documented assumptions by python electronics/generate.py. At the 1 A current limit, a simple linear interpolation from the vendor's 6 V current/torque data estimates 0.0492 N·m per motor, or 6.25 N combined wheel force at 31.5 mm radius. Gearbox losses, current ripple and unit variation reduce confidence in this estimate. It is not a continuous-torque rating. The unloaded 6 V, 250 RPM sample would imply 0.825 m/s wheel speed; the actual 5 V loaded robot will differ. PWM limits are not a speed governor without encoders.

With 75% of nominal 72 Wh treated as useful energy, runtime estimates are 3.6 hours at 15 W average and 2.16 hours at 25 W average. These are planning values. The builder must measure loaded current and an actual discharge cycle to state runtime. Do not describe the result as verified until the physical checks above have passed.

Obtain five required 3766 wheels before accepting wheel and friction-track fit. The previous head servo is removed from the BOM. Recheck current stock and regulator availability before ordering. The firmware and circuit files are checked digitally; head slip setting, loaded turning, noise, servo current and runtime require physical commissioning.

Design research and engineering decisions

This package describes a small, original Dalek-shaped indoor robot. The design uses the requested Adafruit TT motors and orange wheels, an original TTGO T-Display ESP32, a printed shell, two moving arms, and a head that can turn through repeated full revolutions. It is a fabrication design for a first prototype. The CAD and software can be checked digitally. Their operation under actual load still needs the commissioning tests in `assembly.md`.

Source method and date

Product facts were checked against manufacturer product pages, dimensional drawings, component datasheets, and official board source files on 11 September 2026. The required motor and wheel pages were checked first. Component research and exact links are also recorded in `electronics-research.md`, `mechanical.md`, and the BOM CSV files. Prices exclude shipping, tax and regional differences. A product being listed does not guarantee stock when an order is placed.

Quoted electrical ratings describe a specific part and operating condition. Design targets, calculated limits, and assumptions below are identified separately. Neither a successful firmware compile nor a closed STL mesh establishes that a physical robot is safe to drive.

Size and shape

The ROUND-9 design has a circular 300 mm base and remains below 914.4 mm, or three feet. Five complete body sections stack vertically: base, one-piece skirt, shoulder, neck and head. The 213 mm skirt combines the former lower and upper pieces, with the same outer slopes, hemisphere rows and overall robot height. Its middle joint is replaced by an integral rib; four M4 screws, four nuts and eight washers are removed. A common concealed pitch carrier, two decorative arms and a sliding head-motor carriage complete the nine-STL set. The carrier is printed twice. The separate appearance chapter and 100-image source inventory document the retained details. The circular base is a requested customization; photographed bronze television props usually have faceted fenders.

Each body section is designed to print upright on the H2D. Bambu specifies 325 x 320 x 325 mm for single-nozzle printing and 300 x 320 x 325 mm for dual-nozzle printing. The 350 mm combined width is not available to either nozzle individually. The circular base plus an 8 mm brim occupies 316 x 316 mm, so it requires the single-nozzle area and an explicit placement check. The actual exported dimensions and slice placement are recorded for each tested part. Use the supplied print manifest at 100% scale. Scaling changes motor pockets, holes and the stacking registers. [1]

The strong base uses PETG, a 6 mm floor, continuous walls and ribs, and integrated motor pockets. Loads pass through continuous printed material rather than base-quarter bolts or tall motor pillars. The selected structural slice uses six perimeter lines, six top and bottom layers, 40% gyroid infill, and removable support beneath the wheel-well roofs. This geometry and material choice is a design specification, not a measured load rating. The assembly guide specifies a physical proof-load test with the small gearmotors removed from the load path.

The reinforcement has a weight cost. The current `cad/h2d-slice-check.json` separates predicted installed model mass from removable support and brim. Earlier rectangular-body mass values do not describe this revision. Add the battery, all five motors and wheels, four servos, electronics, bearings, fasteners, wires and finish to the current installed-plastic prediction. The original small motors remain as requested; no payload rating is inferred from a successful slice. Their ability to drive and turn the completed assembly must be established with programmed commands and current and temperature measurements on the intended floor.

PLA is suitable for indoor cosmetic body sections, while the base and loaded actuator structures use PETG. The skirt and shoulder are complete rings, with register features that align each tier and fasteners that retain it. Small

functional parts are printed first for fit checks. No separate coupon STL is required. Paint remains clear of the registers, bearing seats and motor pockets.

The requested drive parts set the operating limit

Adafruit 3777 is a low-cost 1:48 plastic gearbox motor. Its allowed supply is 3-6 V. Adafruit reports 1.5 A stalled at 6 V and 0.8 kg.cm stall torque at that voltage. Adafruit 3766 is a 63 mm diameter, 29 mm wide press-fit wheel for the TT shaft. Four separate motors drive four wheels. The wheel page was out of stock when checked; the design retains that exact requested wheel. [2,3]

Stall torque is a brief limiting value, not a continuous operating rating. At the 6 V specification, 0.8 kg.cm is approximately 0.0785 N.m. With a 31.5 mm wheel radius, four motors give an ideal combined stall force of $4 \times 0.0785 / 0.0315 = 9.97$ N. At the chosen 5 V supply and with current limiting, the available force is lower. Gear loss, shaft friction, unequal load and warm windings lower it further. This calculation is an upper bound; it is not a payload rating.

For a design example, a 3 kg robot on a smooth floor with an assumed rolling resistance coefficient of 0.03 requires about 0.88 N just to roll. An acceleration of 0.15 m/s^2 adds 0.45 N. That 1.33 N example is plausible for gentle straight motion but does not include steering scrub. The coefficients and acceleration are engineering assumptions, not measurements of the supplied wheels. Carpet, door thresholds and slopes are outside the initial operating envelope.

With four fixed wheels, steering requires the tires to slide sideways. Moving the wheel mounts inward to fit the circular base reduces the available steering lever arm. Stationary pivot turns cannot be promised on a high-grip floor. Use broad moving arcs first; release the control immediately if a wheel stops. The exact track and wheelbase appear in the mechanical guide. Final track also depends on wheel insertion depth. No earlier steering calculation for the wider rectangular chassis establishes performance of this revision.

Motor speed is also open loop. The motors have no encoders. A PWM percentage controls electrical drive, not a measured speed in metres per second. Motors may need polarity and trim checks. The firmware ramps drive commands and limits their range. Wheel diameter and unloaded motor speed can estimate an upper speed, but loaded speed and stopping distance must be measured on the completed robot. [2]

Driver selection

Three Adafruit DRV8833 boards provide five used H-bridges: four wheel channels and one head-motor channel. The remaining bridge is unused. Each pair of wheel motors on a side receives the same logical direction command. Motor outputs are not wired together. The Adafruit breakout has a nominal 1 A current limit per bridge. Retain its current-sense resistors. A current limit reduces damage during a fault but does not make a stalled motor acceptable for continuous operation. [4]

The DRV8833 operating supply range includes 5 V. The TI datasheet distinguishes continuous, peak and package-dependent current values. The actual breakout's default limit and heat dissipation govern this build. It is not valid to choose a driver merely because a catalogue headline says 2 A peak. Reset-state wiring also matters: the shared sleep line must be low at ESP32 reset, not left to firmware that has not yet started. [4,5]

Battery and power distribution

The selected pack is the Bioenno BLF-1206A, a protected 12 V, 6 Ah LiFePO4 pack with a stated 72 Wh energy rating. The manufacturer lists a 12 A continuous discharge limit, approximately 0.7 kg mass, and the BPC-1502DC 14.6 V, 2 A charger. Its internal protection and large energy reserve are useful for a mobile prop. It supplies converters; it must never connect directly to a 5 V rail, a TT motor, or the T-Display battery socket. [6]

This is the main deliberate exception to an Adafruit-only shopping list. Adafruit's 3.7 V, 6600 mAh pack has a much smaller energy reserve and a stated 3.3 A maximum discharge. Its own page recommends a substantially lower sustained draw. Boosting its output to 5 V while driving five motors and four servos would demand more battery current than the output current. Capacity in mAh alone is not a sufficient selection rule. A protected ready-built pack avoids assembling and balancing loose cells. [7]

Separate regulated 5 V supplies feed the wheel motors, servos, and logic/audio. This prevents a servo pulse or motor start from drawing through the ESP32 board's small power path. All returns meet at the common ground distribution point. The physical actuator switch removes energy from both movement supplies while leaving the controller display available. The main fuse is close to the battery. Wire size, connector ratings, branch fuses and the actual switch DC rating appear in the electrical package.

Runtime is an estimate until measured. With 72 Wh nominal and an assumed 75% usable energy allowance, 54 Wh remain. At an estimated average load of 15 W, that is 3.6 hours; at 25 W, 2.16 hours. These values cover energy only. Temperature, battery condition, low-voltage shutdown, repeated starts and actual servo load change the result. Do not call any of these a guaranteed runtime. Record the first full discharge test with voltage, duration, activity and temperature. [6]

Arms and head

Each arm has two perpendicular positional servo axes. The controller varies the two angles using sine and cosine signals. For small angles, the arm tip follows a near-circular orbit. At large angles, the geometry becomes a spatial curve, so the firmware uses a modest amplitude. Equal pulse amplitudes do not guarantee equal physical angles until the servo centres and directions have been calibrated. The two assemblies can share a phase offset without needing a cam mechanism.

Four MG92B servos provide the arm axes from inside the shoulder. Spherical arm roots cover the sockets; cases, horns and cables remain behind the armor. Their power and signal requirements must both be met: the Adafruit product page specifically calls for 5 V control signals. A PCA9685 running at 3.3 V does not provide a 5 V HIGH just because its separate servo power terminal has 5 V. U8 translates the four servo signals. U9 routes the head motor's 20 kHz PWM under static direction enables. The electrical chapter specifies enable pins, pull-downs, bypassing and unused inputs. [8]

The head uses a fifth Adafruit 3777 TT motor and 3766 wheel. The horizontal tyre contacts the inside of an integral drum beneath the dome. A sliding motor carriage and adjustment screw set contact; four guide clamps with nyloc nuts hold the setting. The tyre supplies compliance. Start with clearance, introduce light contact, lock the setting and check a complete hand revolution for binding and runout. The adjustment is not a calibrated contact-force control. The bearings carry the dome load; the drive wheel supplies tangential force. [2,3]

Head PWM controls drive level and direction, not a measured angle or speed. The design has no head encoder or home sensor. It therefore cannot promise a repeatable face-forward command. Rubber traction, print finish, runout and temperature change the friction behavior. The manual requires loaded rotation and stopping checks at the final adjustment. The controller applies a zero-output interval before reversing the head motor.

The eye and dome decorations contain no moving electrical wires. Audio comes from a speaker fixed in the body. This permits repeated complete rotations without a slip ring. Bearing, shaft, carriage, adjustment and retention hardware is enumerated in the mechanical bill of materials.

Controller and network

The design is for the original ESP32 TTGO T-Display with a 1.14-inch ST7789 display and 4 MB flash. LilyGO publishes the TFT pins, button pins, battery-monitor pins and board drawings. The S3 and other T-Display models have different dimensions and pin assignments and cannot use this firmware and rear mount unchanged. The screen faces outward at the back. Leave its antenna area clear of wires, metal tape and the battery. [10]

ESP32 Wi-Fi supports station mode, access-point mode and their combined use. The robot serves its own browser interface. A nearby phone can connect directly without an Internet connection, or the robot can join an existing 2.4 GHz network. Station credentials are entered locally. No cloud broker or paid service is needed. The exact setup, fallback behavior and displayed addresses are described in `firmware.md`. [11]

Motion needs a current command stream and an explicit arm action. A lost connection, stale command sequence, browser release, physical actuator power loss or safety fault must remove drive commands. Software timeout is an additional control, not a substitute for the physical power switch. The network is local and HTTP is unencrypted; do not expose it through router port forwarding. The Wi-Fi and application password must remain private to the operator.

Voice collection

The package includes 24 newly synthesized MP3s, about 99 seconds total. Their measured file sizes and durations are recorded in `audio/catalog.json`. They use original short transcripts, the installed desktop speech engine, and a reproducible metallic voice effect. They do not contain ripped television clips. The source list is editable and the generation scripts are supplied. All clips are stored in ESP32 flash, so a memory card or network download is unnecessary.

A MAX98357A converts digital audio to speaker power. The selected 8 ohm, 1 W speaker and limited playback gain keep demand within the logic supply budget. The two speaker terminals form a bridge output: neither is a ground terminal. Fix the speaker to the stationary body and route its wires away from the analog voltage-sense leads. Refer to the electrical source notes for amplifier gain and speaker limit calculations.

What is established, and what must be measured

Digital checks establish that the delivered files exist, mesh volumes are positive, exported parts fit the selected print envelope, audio decodes, and firmware compiles. Inspection checks that connector names and pins agree across code and diagrams. These checks are valuable but cannot detect actual servo backlash, moulding variation, printer shrinkage, real traction, bearing friction, battery noise or a loose screw.

The first build must measure fit, loaded mass, wheel clearance, free head travel, arm current, supply sag, temperature and stopping behavior. The assembly guide makes those checks before closing the shell and before the first floor run. If the requested drive parts cannot move the actual mass on the actual surface, the correct response is to reduce mass or change the operating surface. Do not remove current protection or raise motor voltage above its rating to force a result.

References

1. Bambu Lab. H2D technical specifications, single/dual nozzle build volumes.
https://cdn1.bambulab.com/documentation/h2d/en/H2D_Laser_Full_Combo_20250305.pdf and
<https://eu.store.bambulab.com/products/h2d> . Manufacturer PDF indexed text and regional product specifications checked 2026-09-11; direct fetch intermittently unavailable.
2. Adafruit Industries. DC Gearbox Motor - TT Motor, product 3777. <https://www.adafruit.com/product/3777> .
Checked 2026-09-11.

3. Adafruit Industries. Orange and Clear TT Motor Wheel, product 3766. <https://www.adafruit.com/product/3766> . Checked 2026-09-11.
4. Adafruit Learning System. DRV8833 motor driver overview and pinouts. <https://learn.adafruit.com/adafruit-drv8833-dc-stepper-motor-driver-breakout-board/overview> and <https://learn.adafruit.com/adafruit-drv8833-dc-stepper-motor-driver-breakout-board/pinouts> . Guide published 2016, page checked 2026-09-11.
5. Texas Instruments. DRV8833 datasheet, SLVSAR1E, July 2015 revision. <https://www.ti.com/lit/ds/symlink/drv8833.pdf> . Checked 2026-09-11.
6. Bioenno Power. BLF-1206A pack and compatible charger specifications. <https://www.bioennopower.com/products/12v-6ah-lifepo4-battery-pvc> . Checked 2026-09-11.
7. Adafruit Industries. Lithium Ion Battery Pack, product 353. <https://www.adafruit.com/product/353> . Checked 2026-09-11.
8. Adafruit Industries. TowerPro MG92B servo, product 2307. <https://www.adafruit.com/product/2307> . Checked 2026-09-11.
9. Dalek visual source inventory. `docs/reference-image-review.csv` and `docs/appearance-research.md`, 100 reviewed images and the applied appearance decisions, 2026-09-12.
10. LilyGO. TTGO-T-Display original board repository, pinout, schematic and CAD. <https://github.com/Xinyuan-LilyGO/TTGO-T-Display> . Checked 2026-09-11.
11. Espressif Systems. Arduino ESP32 Wi-Fi API. <https://docs.espressif.com/projects/arduino-esp32/en/latest/api/wifi.html> . Current API concepts checked 2026-09-11; the compiled firmware pins the older compatible platform version in `firmware/platformio.ini`.

Dalek appearance reference review

The strongest appearance target is the bronze Time War Dalek: a low dark fender, a tapering body with ordered rows of gold hemispheres, a detailed shoulder band, a dark ringed neck, and a domed head with a raised eye cowl. The revised robot should retain that visual order while using a circular base and fully concealed arm actuators. The circular footprint and adjustable TT-motor head drive are project requirements. They are not claims about the construction of a television prop.

This report defines visual requirements for the mechanical revision. It does not certify the final CAD, printed parts, motor performance, bearing fit, or completed robot. The separate mechanical verification records must establish those points.

Reference inventory and evidence quality

The accompanying reference-image-review.csv (reference-image-review.csv) records 100 selected, visually inspected reference images. Each row contains its source page, original image URL, pixel dimensions, downloaded-byte SHA256, decoded-pixel SHA256, a 256-bit difference hash, an individual observation, and a proposed design implication. Reference IDs D001-D100 are stable within that inventory. All images were accessed and reviewed on 2026-09-12.

The collection contains 50 photographs from an original-prop restoration, 23 bronze-era construction, exhibition and screen images in the Dalek 63•88 archive, 13 official BBC Studios character/story images, 10 BBC America design-history images, one additional original-prop owner photograph, and three official BBC Studios/Propstore bronze auction photographs. These are 100 distinct images, not 100 distinct props. Several images document separate components and successive assembly stages of the same original prop. That coverage is useful for the hidden joints and stacking requirements as well as the exterior finish.

Seventy-seven selected images come from official BBC publications or the owner/restorer of an original prop. Twenty-three come from a specialist historical archive; that archive's historical account is secondary, and original photographer attribution varies. The archive images support visible shape and finish observations, not independent proof of a manufacturing dimension. No dimension in this robot should be justified by counting pixels in an uncalibrated photograph.

The final set has 100 unique image URLs, 100 unique downloaded-byte hashes and 100 unique decoded-pixel hashes. Visual screening also removed repeated crops, non-Dalek character portraits, packaging images, concept collages and weak views. For example, the narrow bronze publicity crop in the archive was omitted in favor of its wider BBC version, and the wide crop of the official execution Dalek was omitted in favor of the larger portrait. Hash uniqueness alone would not establish that those were different photographs. The closest pair among the selected 256-bit difference hashes differed in 89 positions. That calculation is supporting evidence, not a substitute for the visual review.

The reference images and contact sheets remain in temporary analysis storage. They are not included in the repository, the PDF, the video, or the S3 deliverables. The CSV links to the exact originals so the evidence can be inspected without redistributing a third-party image collection.

Overall shape and the circular base

The low fender and broad lower skirt carry much of the recognizable Dalek outline. Official bridge and classic-prop images show the wheels hidden behind the lower body, while the upper body becomes narrower toward the neck. The restored original also shows a small outward ledge below its lowest hemisphere row. These observations support a substantial perimeter bumper, concealed wheels and a clear taper. [D089, BBC bridge photograph](#); [D050, original restoration front view](#).

The photographed bronze props generally have faceted fenders and skirt panels. A perfectly circular base is therefore a deliberate adaptation. It should read as a round version of the same broad fender: a continuous dark lower band, a shallow ledge or bevel above it, and the decorated skirt rising inside that perimeter. A round cylinder with no transition to the body would lose this structure. [D057, bronze full-body view](#).

For this robot, the circular base must be a real geometric boundary in the STL and plan view. Rounded corners on a rectangular base do not meet the requirement. Wheel openings should face downward and inward so they do not create four large exterior notches. The base must still provide independent motor saddles, a continuous structural floor and a clear load path to the stack fasteners. Decorative roundness does not establish strength; the base still needs the mechanical checks and physical commissioning specified elsewhere.

The skirt uses shallow panel relief to organize the hemisphere columns without requiring a polygonal footprint. The combined skirt keeps all four rows aligned within one print. Its remaining stack joints follow the base and shoulder trim lines. Align the front seam with the shoulder before tightening those joints.

Arms and concealed servos

The visible appendage joint is a ball or rounded pivot in a socket surround. An original restoration close-up shows a tube emerging from a ball inside a raised box; another view shows that the surrounding shoulder conceals the rest of the joint. The newer bronze design adds an ornate square plate around a dark circular pivot. [D013, original arm socket](#); [D037, shoulder socket box](#); [D080, official bronze front view](#).

All four arm servo cases, horns, fixing straps and wire loops should sit behind the shoulder skin. The finished exterior should expose only the round socket, the arm stem and the plunger or emitter. A servo-shaped cover on the end of an arm would hide its label but would still spoil the silhouette. The mechanism belongs inside the body. A socket surround integral to the shoulder and a moving spherical cover integral to an existing moving part can preserve the ten-STL limit.

Concealment must hold while the arms move. At every permitted yaw and pitch combination, the socket cover must overlap the shoulder opening. The visible joint must not reveal a servo through an open side or underneath the arm. It must also clear the shoulder, the adjacent arm and the central body panel. The arm stems should remain slim; placing an entire servo in an enlarged external arm shaft would not match the reference.

The proposed circular wiggle needs two coordinated internal axes for each arm. The appearance evidence does not establish the linkage, angle limit or torque. Those are engineering decisions. The operating animation should use the same pivot locations and limits as the CAD and firmware. Show internal mechanisms only during an explicitly marked assembly cutaway, then restore the complete shoulder surface before the exterior operating sequence.

Detail priorities by component

The skirt should retain four ordered levels of hemispheres. Several construction images show that these are repeated shell details, while the bronze examples add dark rims around their roots. Shallow integral annular bezels offer much of that distinction without requiring separate printed rings. Use a consistent diameter within a row and align the columns through the stack joints. The golden dome of each hemisphere should stand out against its darker root and bronze body panel. [D004, skirt construction](#); [D056, nearly finished bronze body](#).

The shoulder benefits most from physical relief. Bronze references show thick rectangular slats with inset faces and round fastener heads. The front central panel is wider than the flanking slats. Side slats are longer where the shoulder offers more height. Their spacing should continue around the back, with only the necessary interruption for the confirmed TTGO T-Display opening. [D058, close slat and appendage photograph](#); [D086, official side photograph](#).

The neck should have three clearly projecting rings over a darker recessed drum. Thin vertical posts and horizontal grille bars give depth to the gaps. Open black space alone can make the neck appear hollow; an unbroken solid bronze drum loses the contrast. A printed dark grille or a dark finish behind integral bars can supply the visual effect. Keep enough service and ventilation access for the actual mechanism. [D015, restored neck structure](#); [D039, neck ring and grille detail](#).

The dome needs a smooth crown, a short lower lip and a raised eye cowl. Shallow panel grooves can run from the cowl region toward the sides. The eye should have a narrow stem, a group of pale concentric discs and a darker larger lens head. A small central recess and several concentric rings give the lens a convincing face. Keep the two dome lamps visibly separate and angled outward. Fine lamp ribbing should be robust enough to survive support removal and handling. [D063, bronze cowl close-up](#); [D100, official lens and lamp close-up](#).

The plunger should have a shallow cup, a defined rim and a smooth stem. The emitter should read as a central tube surrounded by an open cage of rods. Thin movie-prop rods cannot simply be scaled down without regard to print strength. Use fewer or thicker robust rods if needed, but preserve the open cage and muzzle silhouette. Do not add projectile or effect hardware to this decorative component. [D033, assembled cage](#); [D034, muzzle close-up](#).

These details have a clear priority. First preserve the body taper, fender, ringed neck and dome outline. Then add socket covers, hemispheres, slat relief and the eye cowl. Finally add shallow grooves, bolt-head cues and lamp ribs. Small detail should not create more STL files, obstruct a fastener, expose a servo, block airflow or reduce an essential wall thickness.

Color and finish

Use a darker bronze body with warmer gold slats and hemisphere faces. Use near-black in the neck grille, socket balls, hemisphere bezels, plunger cup and lower bumper. Silver appendage stems and cage rods provide a separate material cue. Pale eyestalk discs and a dark lens ring keep the eye legible. These distinctions are directly visible in the official bronze auction side view. [D099, BBC Studios/Propstore side photograph](#).

Use restrained shading in grooves and around bolt heads. Avoid large artificial damage holes or exposed cables. Different studio lighting can make the same bronze prop appear yellow, brown or purple, so the photographs do not justify a single measured paint color. Select actual paints from dry sample pieces under ordinary room light. A decorative blue eye color can suggest the reference without implying that the current head contains a powered lamp. Any functioning light requires a separately documented electrical implementation.

Hidden adjustable friction head drive

The new TT motor and tyre must remain inside the neck and head envelope. Their external visibility is not required to communicate head motion. The prop reference confirms the broader appearance principle: its dome can be lifted separately, and its internal manual control disappears when assembled. [D068, interior dome photograph](#).

The adjustable friction contact is a custom robot mechanism. A service drawing should show the motor, wheel, smooth driven surface, adjustment direction and locking fasteners. The assembled exterior should show none of those details. The adjustment needs a repeatable way to move the tyre into contact and then lock the setting without distorting the neck or forcing the bearing axis off-center. A separate bearing path must carry the dome's weight; tyre pressure should not become the only head support.

The reference review does not determine the correct tyre compression or slip torque. Those depend on the actual Adafruit tyre, printed surface finish, assembled head mass and bearing friction. Commission the mechanism at low power, increase contact only enough for reliable motion, and verify it in both directions. The assembly PDF and video should show how the builder reaches and locks the adjustment after the surrounding parts are fitted. A decorative slot that cannot be reached with a tool does not satisfy adjustability.

Print and assembly implications

Keep each body level as one complete vertically stackable print. Integrate shallow detail into those existing pieces. Preserve a single strong base that fits wholly inside the H2D's usable length and width, including the selected brim and print orientation. A dimensioned plan view must show the new circular outside diameter and the actual wheel envelopes. Slicer checks should be run on the released STL hashes, not on a separate attractive preview model.

Use ordinary purchased fasteners and measured mounting clearances for the hidden hardware. The ten-file limit allows repeated instances of one universal moving part, but the inventory must state both the number of distinct STL designs and the number of physical prints. Avoid using invisible extra printed brackets to make the mechanism work. The video, CAD and assembly instructions must agree about which part carries each servo, head motor and bearing.

Finish recessed neck and socket areas before final stacking. Confirm motor access before installing electronics above the base. Align bump columns and shoulder details before tightening stack bolts. Route the arm leads entirely inside the shoulder, with slack for the permitted motion. Keep the original rear T-Display readable and its controls accessible. The display opening is a functional adaptation and should be a neat rear panel rather than a feature on the front chest.

Visual acceptance for the revised deliverables

The exterior views should prove the intended result from the front, rear, both sides, above and a low angle. A plan view must show a circular base. The wheels should remain concealed in ordinary standing views. The rear view must show the original TTGO display. The shoulders must show only the intended socket surfaces and arm stems at neutral and at the extremes of the complete allowed motion.

An exploded or cutaway view should show both internal arm mechanisms and the head friction drive. A close view should show the adjustment fasteners and tool path. The completed operating view must restore all covers and shoulder walls. Its arm, head and chassis movements must come from the actual revised geometry. No external servo should disappear merely because a camera angle hides it.

Digital visual acceptance is separate from physical proof. The design remains subject to printed fit, supported load, head traction, motor current, heat, clearance and stop-response tests. The report provides a specific and traceable appearance standard; it does not report any of those tests as completed.

Source register

1. The Prop Gallery, [Doctor Who - restoring an original Dalek](#). Primary owner/restoration account; 50 individual restoration photographs selected. Publication date not stated on the page. The CSV links to each full image.
2. Dalek 63•88, [The New Series: Series One](#). Specialist archive; 23 individual bronze-era construction, exhibition and screen images selected. Photographer and copyright holder vary. Publication date not treated as a manufacturing date.
3. BBC Studios, [Daleks - Explore the Whoniverse](#). Official character and story image index; 13 selected images. Each exact original asset URL is recorded.
4. BBC America Editors, [Doctor Who Gallery: The Evolution of Dalek Design](#), 2020-12-31. Ten selected photographs and screen stills; credited BBC.
5. The Prop Gallery, [Doctor Who - Original Dalek](#). One selected close photograph of the restored original prop. Publication date not stated on the page.
6. BBC Studios / Propstore, [Everything available in the Doctor Who x Propstore auction](#), 2026 auction coverage. Three selected photographs of the bronze prop, including front, side and lens detail.

All source pages and individual image assets were accessed on 2026-09-12. Direct visible observations are distinguished from proposed design implications in the CSV. The reference count describes the reviewed inventory and does not imply permission to redistribute its copyrighted contents.

Original robot voice collection

The 24 MP3 files are in `../firmware/data/audio/`. They are included there so the same files can be uploaded to the robot's LittleFS flash filesystem. `catalog.json` records each exact transcript, duration, size and SHA-256. `phrases.csv` is the editable source list.

These are newly synthesized robot lines, not audio extracted from Doctor Who, a BBC recording, or a particular actor's performance. The text was written for this build. The source is the installed Microsoft David Desktop voice through Windows System.Speech. A low oscillator adds a metallic sound. The collection includes brief familiar words such as "Exterminate" alongside original control and character phrases. No downloaded sound samples are used.

The supplied format is mono MP3, 22,050 Hz, 32 kbit/s. Playback uses the ESP32 and the MAX98357A amplifier in the body. The speaker and all its wires stay stationary during head rotation. The controller lists files automatically. The files are manual sound buttons; the presence of a warning phrase does not mean firmware automatically plays that warning.

To recreate the collection on this Windows machine:

```
powershell -NoProfile -ExecutionPolicy Bypass -File C:/dev/robots/dalek/scripts/generate_voices.ps1
```

This requires the installed Microsoft desktop voice, Python with NumPy, and FFmpeg/FFprobe on PATH. It runs once. It registers no scheduled task. It creates a unique temporary WAV directory and removes only that directory when finished. The persistent outputs are the MP3s and catalog.

To replace a line, edit its row in `phrases.csv`, regenerate, rebuild the filesystem with PlatformIO, and upload that filesystem while actuator power is off. The waveform effect uses a 0.86 playback-rate multiplier, nonlinear amplitude compression, 30 Hz ring modulation, a quiet 45 ms echo, and a 170-3400 Hz frequency band. Peak amplitude is limited before MP3 encoding. Start with amplifier volume low and check the speaker for distortion.

To add your own recordings, use lowercase filenames with letters, numbers, underscores or hyphens and the `.mp3` suffix. Keep the entire flash filesystem inside its partition. Use only recordings you have permission to use. Character names and third-party voice/tool software retain their respective ownership; this package does not grant rights in third-party material.

Digital verification and physical limits

ROUND-9 combines the lower and upper skirts into one 213 mm print. The robot remains nominally 563.8 mm tall. The revision removes one printed piece and one bolted interface; it does not establish a measured increase in strength. Physical fit, adhesion, base strength, loaded driving and stopping still require the commissioning gates in `assembly.md`.

Reference review and preserved behavior

The existing 100-image reference inventory and appearance review remain applicable. Each source has its original URL, dimensions, file/pixel hashes, observation and design implication. No reference image has been redistributed with the package.

Eight other STL designs are byte-identical to ROUND-10. They account for nine physical prints because the pitch carrier is used twice. Firmware, circuits, original MP3s, the four ground drives, rear T-Display, concealed arm mechanisms and adjustable head friction drive are unchanged. PCB mounting hardware and access instructions are updated for the deeper shell.

Printable geometry

Command:

```
python scripts/export_cad.py --check-only
```

Result: nine watertight, consistently wound STL designs, each one connected positive solid, for ten printed pieces. The total solid-material upper bound is 2995.9 g. This is a mesh-volume calculation, not measured or sliced print mass.

The new `02_skirt.stl` is 300.331848 x 300.331848 x 213 mm. Its SHA256 is `ff1baa0536e36c0730d69f5cf8004e069f4fe4e430026d6f3e6bc734a125146e`. All 39,728 faces are nondegenerate and every edge has two adjacent faces. The final mesh replaced a rejected export with a microscopic internal surface at the old rib-cone junction. A single revolved polygon now forms the reinforcing rib.

The old lower and upper skirt STLs are removed. The new part retains both outer slopes, all four hemisphere rows, trim and the visible middle band. It retains the base and shoulder interfaces but replaces the middle bolted flanges with an integral tapered rib. Comparison of 8,496 exterior rays against the former pair differed by at most 0.000004554 mm. The other eight STL hashes are unchanged.

The 213 mm part is below the 320 mm maximum-minus-5 mm height target. Its requested 5 mm brim allowance also fits the H2D footprint. The full printer envelope is not stretched or scaled to obtain this result. See `cad/validation.json` and `bom/printed-parts.csv`.

Mechanical and assembly access checks

Commands:

```
python scripts/check_mechanical.py
python scripts/check_skirt_access.py
```

The complete current mechanical run passed 118 checks across 554,129 samples against the nine current STL hashes. This includes merged-skirt height, interfaces, bore and long-driver checks, unchanged mechanisms and 4,800 current-body sight rays with no visible servo targets. These finite samples are not an exact continuous collision proof. Results and limits are in `cad/mechanical-checks.json` and `docs/revision-review.md`.

The focused access report passes 61 checks across 868,363 sample evaluations. Twenty affected plate/retained-hardware checks were rerun after the final insertion-height change; 41 unchanged tool and battery checks were retained only after their mesh hashes matched. The final incremental command was `python scripts/check_skirt_access.py --plates-only`. A full run remains available. See `cad/skirt-access-checks.json`.

Access relies on the specified 250 mm driver, crowfoot, universal joint and extension. The crowfoot holds the M4 nut while the screw turns. The M2.5 plate nuts require a 5 mm socket. Actual tool profiles must fit the documented conservative envelopes; a tool name alone does not prove clearance.

Retained PCB nuts, washers and adhesive must project no more than 3 mm below the FR4. The plate underside travels at base-local Z44 before moving outward, then lowers to Z34. All underside fittings must remain within the plate outline and wholly outside the 8 mm-radius keepouts around the eight plate-to-base mounting axes. Actual PCB hole locations are a physical template-fit gate. The checker does not invent mounting-hole coordinates from board outlines or certify arbitrary populated plates.

Boards are fitted from above after the plates are seated, with the battery absent. Fasteners must match the actual board holes. The Pololu 4091 regulators have three M2 mounting holes per board. Pre-retained nuts and measured screw lengths avoid inaccessible underside work. Verify the specified motor-case clearance, thread engagement, absence of trace contact and service access before final assembly.

The separate eight arm-order checks across 187,920 samples are retained from ROUND-10. Their four relevant STL hashes are unchanged. `cad/arm-assembly-checks.json` preserves that provenance. The required order still feeds and supports the arms before installing pitch-servo carriers; horns and opaque fabric liners follow.

Actual H2D slicing

Command:

```
python scripts/slice_h2d.py --parts 02_skirt
```

The merged skirt passed an actual Bambu Studio 02.08.02.61 slice using the installed H2D 0.4 mm machine profile in isolated temporary state. Supported assembly-list input places it at X162.5,Y160 without changing the printer/nozzle geometry. Exact triangle count, rigid transform, full 213 mm deposited height, material, process settings and all deposited model/support/brim paths were checked.

The new skirt's bead bounds are X7.7093..317.2917 and Y5.2093..314.7917 mm, within the left 325 x 320 mm area. It predicts 527.879 g of installed model and about 1033.87 g total PLA including supports and brim, with a 23 hour 45 minute 40 second print time. One 1 kg spool is insufficient; allow at least 1.14 kg for this job with reserve and plan filament capacity or a compatible change procedure.

The release report combines this new slice with eight unchanged slice records whose exact STL hashes were checked. It covers nine designs and ten physical prints:

- Installed printed model: 2345.25 g.
- Total filament including support and brim: 3951.62 g.
- Sequential print time: 406685.32 seconds, about 112.97 hours.
- Compared with ROUND-10: 83.95 g less installed plastic, 121.44 g less filament and 3 hours 18 minutes 57 seconds less predicted print time.

The BOM keeps 3 kg each of PETG and PLA as supply allowances, including reserve for the long skirt print. These quantities are not claimed deposited mass. Bambu omitted a separate Brim feature on some retained supported parts despite the requested brim setting; that behavior remains explicitly recorded. All actual generated deposition

fits. Inspect adhesion paths before printing.

Controls, wiring and media

The unchanged firmware and filesystem builds retain their prior passing results: RAM 45,416 bytes, application flash 995,025 bytes and a 2,097,152-byte LittleFS image. Native controls and browser tests already cover the 8-degree arm limit, head PWM cap, reversal delay, fault stop, command leases and no automatic restart. These checks were not needlessly rerun for an unchanged electrical design. All 24 original MP3 hashes remain unchanged. The wiring schedule remains 185 rows with four SVG circuit sheets.

The drawing package now has 15 PNGs: six overview/mechanism sheets and nine component drawings. Five additional CAD previews serve the PDF. Their manifest verifies current source hashes and ZIP contents. The video shows one skirt installation and the plate-first, battery-last sequence, with nine designs and ten prints. Full FFmpeg decoding and FFprobe validate 120 seconds, 2880 frames, 1920 x 1080, 24 fps and AAC audio. Operation is labelled as simulation.

The PDF is rendered and visually reviewed before delivery. The existing GitHub Actions/OIDC workflow publishes the committed PDF and video to private S3 storage and verifies downloaded file hashes. The final email contains the PDF and a signed video link with an explicit expiry. Workflow and SES acceptance identifiers are recorded in the execution plan.

Physical limits

No robot was printed, wired, powered or driven during this revision. The predicted 2.345 kg of installed plastic excludes the battery, motors, wheels, servos, electronics, fasteners, fabric and finish. Weigh the finished robot and complete the physical checks. A continuous printed skirt and fewer joints do not by themselves prove higher strength, reliable traction or a payload rating.

Electrical bill of materials

bom\electronics.csv

- 1. reference:** M1-M5 | **quantity:** 5 | **item:** TT geared motor for four wheels and head friction drive | **manufacturer part:** Adafruit 3777 | **unit** **usd:** 2.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock on2026-09-12 | **url:** <https://www.adafruit.com/product/3777> | **notes:** Required motor;3-6V operated at5V;each uses its own1A limited bridge
- 2. reference:** W1-W5 | **quantity:** 5 | **item:** Orange clear TT wheel | **manufacturer part:** Adafruit 3766 | **unit** **usd:** 1.50 | **price basis:** listed | **availability checked 2026 09 11:** Out of stock on2026-09-11;recheck before purchase | **url:** <https://www.adafruit.com/product/3766> | **notes:** Four ground wheels plus one head friction wheel;63x29mm;obtain exact part before fit signoff
- 3. reference:** U1 | **quantity:** 1 | **item:** Original TTGO T-Display ESP32 1.14 inch | **manufacturer part:** LILYGO TTGO T-Display | **unit** **usd:** 25.00 | **price basis:** allowance | **availability checked 2026 09 11:** Exact revision and stock must be checked | **url:** <https://github.com/Xinyuan-LilyGO/TTGO-T-Display> | **notes:** Existing user board can remove this cost; not T-Display-S3 or AMOLED
- 4. reference:** U5-U6 U11 | **quantity:** 3 | **item:** DRV8833 dual motor driver | **manufacturer part:** Adafruit 3297 | **unit** **usd:** 5.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock on2026-09-12 | **url:** <https://www.adafruit.com/product/3297> | **notes:** Keep stock0.2ohm sense resistors;U5/U6 four wheel bridges;U11A head;U11B unused
- 5. reference:** U10 | **quantity:** 1 | **item:** PCA9685 servo PWM board | **manufacturer part:** Adafruit 815 | **unit** **usd:** 14.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/815> | **notes:** Remove OE pull-down; VCC3.3V; leave V+ unconnected
- 6. reference:** SV0-SV3 | **quantity:** 4 | **item:** MG92B positional metal gear micro servo | **manufacturer part:** Adafruit 2307 | **unit** **usd:** 11.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/2307> | **notes:** Includes horns and horn screws; needs 5V control signals
- 7. reference:** U8-U9 | **quantity:** 2 | **item:** 74AHCT125 DIP14 level buffer | **manufacturer part:** Adafruit 1787 | **unit** **usd:** 1.50 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/1787> | **notes:** U8 four5V servo signals;U9 headPWM routing;useAHCT powered-off-input tolerance;verify actual IC marking
- 8. reference:** U2-U3 | **quantity:** 2 | **item:** 5V5.5A buck regulator | **manufacturer part:** Pololu D36V50F5 4091 | **unit** **usd:** 39.95 | **price basis:** listed | **availability checked 2026 09 11:** Rationed active and preferred; backorders allowed | **url:** <https://www.pololu.com/product/4091> | **notes:** Separate motor and servo regulators;25.4x25.4x9.5mm; no outputs paralleled
- 9. reference:** U4 | **quantity:** 1 | **item:** 5V1.2A logic buck regulator | **manufacturer part:** Adafruit MPM3610 4739 | **unit** **usd:** 6.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/4739> | **notes:** Input max21V; solder power pins; EN left open
- 10. reference:** B1 | **quantity:** 1 | **item:** Protected12V6Ah72WhLiFePO4 battery | **manufacturer part:** Bioenno BLF-1206A | **unit** **usd:** 79.99 | **price basis:** listed | **availability checked 2026 09 11:** Add to cart offered | **url:** <https://www.bioennopower.com/products/12v-6ah-lifepo4-battery-pvc> | **notes:** 12A continuous;0.7kg;PP30 discharge;rotated padded70X114Y76Z envelope;76X116Y pocket;connector notchleft
- 11. reference:** CH1 | **quantity:** 1 | **item:** Matched14.6V2A LiFePO4 charger | **manufacturer part:** Bioenno BPC-1502DC | **unit** **usd:** 21.99 | **price basis:** listed | **availability checked 2026 09 11:** Add to cart offered | **url:** <https://www.bioennopower.com/products/lithium-12v-2a-amp-lifepo4-battery-charger> | **notes:** Use factory battery charge connector; unplug and remove battery to charge
- 12. reference:** U7 | **quantity:** 1 | **item:** MAX98357A mono I2S amplifier | **manufacturer part:** Adafruit 3006 | **unit** **usd:** 5.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/3006> | **notes:** GAIN100k to5V gives3dB; fixed digital gain0.28
- 13. reference:** SPK1 | **quantity:** 1 | **item:** 8ohm1W3inch speaker | **manufacturer part:** Adafruit 1313 | **unit** **usd:** 1.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/1313> | **notes:** 77.8x77.8x25.49mm; never ground either speaker lead
- 14. reference:** S1-S2 | **quantity:** 2 | **item:** Panel SPST ON-OFF disconnect | **manufacturer part:** NKK S1A | **unit** **usd:** 12.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** https://www.nkkswitches.com/wp-content/themes/impress-blank/search/inc/part.php?part_no=S1A | **notes:** 20A30VDC;12.5mm hole; includes M12 panel hardware; OFF label required
- 15. reference:** FH1-FH4 | **quantity:** 4 | **item:** Inline ATO fuse holder16AWG | **manufacturer part:** Littelfuse 0FHA0001ZXJ | **unit** **usd:** 4.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:**

<https://info.littelfuse.com/ato-fha-series-landing-page> | **notes:** 20A holder; insulate leads and restrain wiring

16. reference: F1 | **quantity:** 1 | **item:** ATO7.5A32V fuse | **manufacturer part:** Littelfuse 025707.5 | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** <https://www.littelfuse.com/products/fuses-overcurrent-protection/fuses/automotive-passenger-car/blade-fuses> | **notes:** Main pack fuse within 100mm of connector

17. reference: F2-F3 | **quantity:** 2 | **item:** ATO3A32V fuse | **manufacturer part:** Littelfuse 0257003 | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** <https://www.littelfuse.com/products/fuses-overcurrent-protection/fuses/automotive-passenger-car/blade-fuses> | **notes:** Input branch fuses for motor and servo converters

18. reference: F4 | **quantity:** 1 | **item:** ATO1A32V fuse | **manufacturer part:** Littelfuse 0257001 | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** <https://www.littelfuse.com/products/fuses-overcurrent-protection/fuses/automotive-passenger-car/blade-fuses> | **notes:** Input fuse for logic converter

19. reference: PP30 | **quantity:** 1 | **item:** Anderson Powerpole mating red-black pair with contacts | **manufacturer part:** PP30 genuine housings and 30A contacts | **unit usd:** 6.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** <https://www.andersonpower.com/us/en/resources/PowerpoleResourcesPage.html> | **notes:** Use contacts matched to 18AWG; do not cut factory battery lead

20. reference: PROTO | **quantity:** 2 | **item:** Half-size solderable prototype board | **manufacturer part:** Adafruit 1609 | **unit usd:** 4.50 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/1609> | **notes:** One buffer board; one low-current divider board; power distribution uses separate heavy wire

21. reference: SOCKET | **quantity:** 2 | **item:** DIP14 IC socket | **manufacturer part:** 14pin0.3inch socket | **unit usd:** 0.75 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Protects buffers during soldering

22. reference: R1-R8 R12 R20-R24 R40-R45 | **quantity:** 20 | **item:** 10kohm1percent0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Input pull-downs; PCA OE pull-up; head PWM pull-down; two headgate pull-ups

23. reference: R10 R14 | **quantity:** 2 | **item:** 100kohm1percent0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Battery divider top and amplifier gain resistor

24. reference: R11 | **quantity:** 1 | **item:** 22kohm1percent0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Battery divider bottom; never replace with 33k

25. reference: R13 | **quantity:** 1 | **item:** 15kohm1percent0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Actuator sense bottom resistor

26. reference: R30-R35 | **quantity:** 6 | **item:** 220ohm0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Four servo signal and two head bridge input series resistors

27. reference: R15-R16 | **quantity:** 2 | **item:** 100ohm1W resistor | **manufacturer part:** Metal film flameproof axial | **unit usd:** 0.40 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Actuator rail discharge; dissipates 0.25W each during use

28. reference: C8-C11 MOTOR_CAPS | **quantity:** 9 | **item:** 100nF25V or higher ceramic capacitor | **manufacturer part:** X7R radial | **unit usd:** 0.15 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Five motor suppression; two buffer bypass; two ADC filters

29. reference: C1-C2 | **quantity:** 2 | **item:** 220uF25V lowESR capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 0.75 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Motor and servo converter input; polarity critical

30. reference: C3 | **quantity:** 1 | **item:** 100uF25V capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 0.50 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Logic converter input

31. reference: C4-C5 C7 C12 | **quantity:** 4 | **item:** 470uF10V lowESR capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 0.75 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** One at each of three drivers and one at logic output

- 32. reference:** C6 | **quantity:** 1 | **item:** 1000uF10V lowESR capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Servo distribution block; polarity critical
- 33. reference:** POWER_WIRE | **quantity:** 4 | **item:** 18AWG flexible stranded copper wire metre | **manufacturer part:** Silicone insulation red-black-orange | **unit usd:** 1.50 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Combined total 4m; trim to actual path plus service loop
- 34. reference:** BRANCH_WIRE | **quantity:** 5 | **item:** 22AWG flexible stranded wire metre | **manufacturer part:** Red and black copper | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Individual servo and logic branches
- 35. reference:** SIGNAL_WIRE | **quantity:** 5 | **item:** 26AWG stranded signal wire metre | **manufacturer part:** Assorted color copper | **unit usd:** 0.60 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Keep I2C and I2S runs short under 200mm where possible
- 36. reference:** CONNECTORS | **quantity:** 1 | **item:** Connector and termination allowance | **manufacturer part:** Insulated terminal blocks and ferrules and servo header extensions | **unit usd:** 18.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Power blocks rated $\geq 10A30VDC$; four servo 3pin sockets; removable run-power link; five 2pin motor plugs
- 37. reference:** INSULATION | **quantity:** 1 | **item:** Electrical assembly consumables | **manufacturer part:** Heatshrink cable ties labels solder and foam | **unit usd:** 12.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Insulation and strain relief are required

Mechanical hardware bill of materials

bom\hardware.csv

- 1. category:** fastener | **item:** M4x20 socket screw | **quantity:** 12 | **unit:** piece | **location:** Three stack interfaces | **specification:** Four per interface;4.4mm clearance holes; no middle skirt joint
- 2. category:** fastener | **item:** M4 nyloc nut | **quantity:** 12 | **unit:** piece | **location:** Stack interfaces | **specification:** Use7mm long-reach nut tool below base flange
- 3. category:** fastener | **item:** M4 plain washer | **quantity:** 24 | **unit:** piece | **location:** Stack interfaces | **specification:** 8-9mmOD; one below each head and nut
- 4. category:** fastener | **item:** M3x40 socket screw | **quantity:** 16 | **unit:** piece | **location:** Motor clamps | **specification:** Four per motor; through29mm boss and floor
- 5. category:** fastener | **item:** M3 hex nut | **quantity:** 25 | **unit:** piece | **location:** Motor clamps speaker display jackscrew | **specification:** 16 motor plus4 speaker plus4 display plus1 captured jackscrew nut
- 6. category:** fastener | **item:** M3 nyloc nut | **quantity:** 4 | **unit:** piece | **location:** Head slide clamps | **specification:** Locks four guide screws after setting contact; no separate jackscrew jam nut
- 7. category:** fastener | **item:** M3 plain washer | **quantity:** 52 | **unit:** piece | **location:** Motor clamps speaker display slide underside | **specification:** 32 motor plus8 speaker plus8 display plus4 slide underside
- 8. category:** retainer | **item:** Stainless clamp strip43x8x1mm | **quantity:** 8 | **unit:** piece | **location:** Two per motor | **specification:** Two3.4mm holes37mm apart;3mm from ends; deburr and pad
- 9. category:** fastener | **item:** M3x12 socket screw | **quantity:** 4 | **unit:** piece | **location:** Speaker | **specification:** 4mm baffle plus speaker tab washers and nut; verify actual flange
- 10. category:** fastener | **item:** M3x25 socket screw | **quantity:** 4 | **unit:** piece | **location:** Display retaining tabs | **specification:** Through integrated16mm frame
- 11. category:** retainer | **item:** Nonconductive retaining tab20x8x1mm | **quantity:** 4 | **unit:** piece | **location:** Display | **specification:** Polycarbonate or nylon strip;3.4mm hole3mm from outer end;17mm inward reach; pad bare PCB edge
- 12. category:** fastener | **item:** M3x8 plastic thread-forming screw | **quantity:** 6 | **unit:** piece | **location:** Bearing retainers | **specification:** Three top and three bottom;2.6mm pilots
- 13. category:** retainer | **item:** M3 steel washer9mmOD1mmthick | **quantity:** 6 | **unit:** piece | **location:** Bearing retainers | **specification:** At14mm screw radius; overlap outer race only
- 14. category:** fastener | **item:** M8x70 hex-head bolt | **quantity:** 1 | **unit:** piece | **location:** Head spindle | **specification:** Must pass8mm bearing bore freely
- 15. category:** fastener | **item:** M8 nyloc nut | **quantity:** 1 | **unit:** piece | **location:** Head spindle | **specification:** 13mm socket via24mm crown hole
- 16. category:** fastener | **item:** M8 washer16mmOD | **quantity:** 1 | **unit:** piece | **location:** Head hub top | **specification:** Only above printed hub
- 17. category:** bearing | **item:** 608-2RS bearing | **quantity:** 2 | **unit:** piece | **location:** Neck tower | **specification:** 22mmOD8mmID7mmwidth
- 18. category:** spacer | **item:** Metal spacer8.1x12x12mm | **quantity:** 1 | **unit:** piece | **location:** Between bearing inner races | **specification:** 8.1mmID12mmOD12mm long; square parallel ends
- 19. category:** spacer | **item:** Precision shim8x12x1mm | **quantity:** 2 | **unit:** piece | **location:** Bearing inner races | **specification:** One below and one above; narrow OD essential
- 20. category:** fastener | **item:** M3x16 socket screw | **quantity:** 4 | **unit:** piece | **location:** Head radial slide | **specification:** Through3mm carriage and4mm neck deck; clamp after adjustment
- 21. category:** fastener | **item:** M3 washer6mmOD1mmthick | **quantity:** 4 | **unit:** piece | **location:** Head slide screw heads | **specification:** Small OD clears motor-saddle walls; do not substitute wide washers

22. category: fastener | **item:** M3x25 socket screw | **quantity:** 1 | **unit:** piece | **location:** Head contact jackscrew | **specification:** Through captured M3 nut; four slide clamps lock the chosen setting

23. category: fastener | **item:** M2x10 machine screw | **quantity:** 8 | **unit:** piece | **location:** Four arm servo horns | **specification:** Two each plus supplied spline screws

24. category: fastener | **item:** M2x14 machine screw | **quantity:** 6 | **unit:** piece | **location:** Two Pololu4091 regulators | **specification:** Three per board; trim to measured working length for at least2mm motor-case gap and two exposed threads; never forceM2.5 throughM2 holes

25. category: fastener | **item:** M2 hex nut maximum1.6mm thick | **quantity:** 14 | **unit:** piece | **location:** Arm horns and Pololu regulators | **specification:** Eight for horns plus six pre-retained below regulator mounts

26. category: fastener | **item:** M2 plain washer0.3mm thick | **quantity:** 34 | **unit:** piece | **location:** Arm horns and Pololu regulators | **specification:** 16 for horns plus three per regulator screw; two under PCB screwhead and one below FR4

27. category: panel | **item:** Unclad FR4 plate160x56x2mm | **quantity:** 2 | **unit:** piece | **location:** Electronics mounting | **specification:** Purchased sheet cut and drilled; X +/-80 Y +/-93 centres; not an extra printed part

28. category: fastener | **item:** M2.5x40 machine screw | **quantity:** 8 | **unit:** piece | **location:** FR4 plates to base | **specification:** Four per plate through9mm mounting pad25mm spacer2mm plate; verify two threads beyond nut

29. category: spacer | **item:** Nylon spacer25mm long M2.5 bore | **quantity:** 8 | **unit:** piece | **location:** Electronics plates | **specification:** Plate underside localZ34 clears ribs and motors

30. category: fastener | **item:** M2.5x14 machine screw | **quantity:** 32 | **unit:** piece | **location:** Electronics boards | **specification:** Insert from above into pre-retained underside nuts; trim16mm if needed; verify at least2mm motor-case gap and two full exposed threads

31. category: spacer | **item:** Nylon spacer6mm long M2.5 bore | **quantity:** 32 | **unit:** piece | **location:** Electronics boards | **specification:** Clearance above FR4; unused pieces are spares

32. category: fastener | **item:** M2.5 standard hex nut | **quantity:** 8 | **unit:** piece | **location:** FR4 plate mounts | **specification:** Eight base-to-plate joints;5mm wrench size

33. category: fastener | **item:** M2.5 thin hex nut maximum1.6mm thick | **quantity:** 32 | **unit:** piece | **location:** Retained underside PCB nuts | **specification:** DIN439 or ISO4035;5mm wrench size; pre-retain outside8mm plate-mount keepouts; unused pieces are spares

34. category: fastener | **item:** M2.5 plain washer0.5mm thick | **quantity:** 112 | **unit:** piece | **location:** Electronics | **specification:** Two under each of32 PCB screwheads plus one below each PCB plate joint; eight base-to-plate joints use one each head and nut; unused pieces are spares

35. category: retainer | **item:** 20mm hook-and-loop strap350mm long | **quantity:** 2 | **unit:** piece | **location:** Battery | **specification:** Independent straps through integral slots

36. category: retainer | **item:** 2.5mm cable tie200mm long | **quantity:** 40 | **unit:** piece | **location:** Servos head TT motor and wiring | **specification:** 8 arm-servo ties2 head-motor ties; remainder routing and spares

37. category: padding | **item:** EPDM foam sheet300x200mm | **quantity:** 1 | **unit:** set | **location:** Motors battery servos display | **specification:** Nonconductive1mm2mm3mm assortment; keep off hot components

38. category: consumable | **item:** Removable thread-locking compound | **quantity:** 1 | **unit:** bottle | **location:** Metal-to-metal only | **specification:** Keep off plastic bearings and nyloc nuts

39. category: consumable | **item:** Nonconductive epoxy | **quantity:** 1 | **unit:** set | **location:** Retained underside PCB nuts | **specification:** Small side fillets hold nut and washer for assembly only; cured adhesive plus washer and nut must project no more than3mm below FR4; keep threads and PCB clear

40. category: consumable | **item:** PETG filament | **quantity:** 3 | **unit:** kg | **location:** Structural prints | **specification:** Current model and support estimate plus10percent allowance rounded up to1kg spools

41. category: consumable | **item:** PLA filament | **quantity:** 3 | **unit:** kg | **location:** Cosmetic prints | **specification:** Supply allowance retained for model and supports plus a large-skirt retry contingency; not predicted consumption

42. category: consumable | **item:** Paint and finish supplies | **quantity:** 1 | **unit:** set | **location:** Integrated details | **specification:** Paint bumps eye and lamps; no separate decoration meshes

43. category: accessory | **item:** Opaque removable tape30x30mm | **quantity:** 1 | **unit:** piece | **location:** Optional crown cover | **specification:** Remove for spindle service; no printed cap

- 44. category:** accessory | **item:** Acoustic cloth100x100mm | **quantity:** 1 | **unit:** piece | **location:** Speaker grille guard | **specification:** Attach inside central front grille without touching cone
- 45. category:** accessory | **item:** Opaque black stretch fabric80x80mm | **quantity:** 2 | **unit:** piece | **location:** Hidden arm socket liners | **specification:** 44mm centre opening and overlapped radial slit; bond inside spherical-root rear rim and fixed socket; retain loose folds
- 46. category:** accessory | **item:** Flexible fabric adhesive | **quantity:** 1 | **unit:** tube | **location:** Socket liners and acoustic cloth | **specification:** Compatible with printed plastic and fabric; keep away from moving horns
- 47. category:** padding | **item:** Nonconductive1mm tab shim | **quantity:** 4 | **unit:** piece | **location:** Speaker | **specification:** Use only if surround clearance requires; cut from purchased sheet
- 48. category:** retainer | **item:** Hook-and-loop cable strap | **quantity:** 6 | **unit:** piece | **location:** Service harness | **specification:** Disconnectable loops at body interfaces
- 49. category:** tool | **item:** 250mm3mm ball-end hex driver rated20degrees | **quantity:** 1 | **unit:** piece | **location:** Base to one-piece skirt bolts | **specification:** 16degree inward approach through184mm top opening;250mm exposed shaft required
- 50. category:** tool | **item:** 2.5mm L-hex key short leg15mm | **quantity:** 1 | **unit:** piece | **location:** Head slide clamp screws | **specification:** Short leg may be cut and deburred; horizontal shaft passes neck-localZ49.9 through service ports
- 51. category:** tool | **item:** Bahco677-7 crowfoot7mm1/4drive | **quantity:** 1 | **unit:** piece | **location:** M4 base to skirt nuts | **specification:** 5mm jaw thickness22mm width17.5mm drive offset; inward drive axisR119.5; verify actual boss and jaw envelope
- 52. category:** tool | **item:** Bahco6966 universal joint1/4drive | **quantity:** 1 | **unit:** piece | **location:** Deep stack and plate nut access | **specification:** 35mm long13.9mm OD; angled extension through184mm top bore
- 53. category:** tool | **item:** Bahco6960-11 extension1/4drive | **quantity:** 1 | **unit:** piece | **location:** Deep stack and plate nut access | **specification:** At least250mm reach; nominal275mm listing; shaft8mm and socket12mm diameter
- 54. category:** tool | **item:** 5mm socket1/4drive | **quantity:** 1 | **unit:** piece | **location:** M2.5 plate and board nuts | **specification:** Maximum14mm OD combined with universal joint; install plate screws from below; battery and tall boards absent
- 55. category:** tool | **item:** Metric hex keys and nut drivers | **quantity:** 1 | **unit:** set | **location:** Remaining fasteners | **specification:** 13mm socket outside diameter<=23mm for head access
- 56. category:** tool | **item:** Caliper and2.1mm2.8mm3.4mm drills | **quantity:** 1 | **unit:** set | **location:** Fit and hardware preparation | **specification:** Measure actual parts; deburr carefully
- 57. category:** tool | **item:** Digital scale and DC current meter | **quantity:** 1 | **unit:** set | **location:** Commissioning | **specification:** Measure finished mass and drive load

Printed part quantities and orientations

bom\printed-parts.csv

1. part: 01_base | **quantity:** 1 | **material:** PETG | **x mm:** 300.0 | **y mm:** 300.0 | **z mm:** 57.0 | **volume mm3:** 745898.54 | **solid mass g:** 947.29 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 8 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Floor down; single nozzle | **notes:** Diameter300 circular unibody;6mm floor;4mm wall/ribs;6walls6top/bottom40%gyroid;8mmbrim;normal auto supports below wheel-well roofs;remove through bottom wells

2. part: 02_skirt | **quantity:** 1 | **material:** PLA | **x mm:** 300.332 | **y mm:** 300.332 | **z mm:** 213.0 | **volume mm3:** 518836.51 | **solid mass g:** 643.36 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Large opening down | **notes:** One complete210mm two-slope body plus3mm registry; four hemisphere rows and integral tapered middle rib; no middle bolted joint;4walls15%infill;normal auto supports

3. part: 04_shoulder | **quantity:** 1 | **material:** PETG | **x mm:** 226.0 | **y mm:** 230.2 | **z mm:** 123.0 | **volume mm3:** 552905.89 | **solid mass g:** 702.19 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Bottom flange down | **notes:** Concealed servo gunboxes spherical sockets slats rear display frame; normal auto supports inside gunboxes;4walls25%infill

4. part: 05_neck | **quantity:** 1 | **material:** PETG | **x mm:** 220.0 | **y mm:** 220.0 | **z mm:** 66.0 | **volume mm3:** 210862.12 | **solid mass g:** 267.79 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Bottom flange down | **notes:** Integrated three rings grille bearing tower radial slide deck jackscrew lug; support deck underside;4walls30%infill

5. part: 06_head | **quantity:** 1 | **material:** PLA | **x mm:** 220.0 | **y mm:** 285.667 | **z mm:** 114.025 | **volume mm3:** 254128.48 | **solid mass g:** 315.12 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Drum rim down | **notes:** Dome eye discs tilted lamps hub friction drum spokes integrated; normal auto supports inside dome under eye and hub;4walls15%infill

6. part: 07_pitch_carrier | **quantity:** 2 | **material:** PETG | **x mm:** 49.0 | **y mm:** 30.0 | **z mm:** 49.0 | **volume mm3:** 11843.93 | **solid mass g:** 15.04 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Floor down | **notes:** Print this same file twice;4walls30%infill; actual horn and servo fit gate

7. part: 08_plunger_arm | **quantity:** 1 | **material:** PLA | **x mm:** 138.0 | **y mm:** 50.447 | **z mm:** 50.593 | **volume mm3:** 25870.08 | **solid mass g:** 32.08 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** As exported; ensure on bed | **notes:** Spherical root and telescoping rings integrated; support cup and root; supplied servo horn; decorative arm only

8. part: 09_emitter_arm | **quantity:** 1 | **material:** PLA | **x mm:** 139.0 | **y mm:** 50.447 | **z mm:** 50.593 | **volume mm3:** 24255.99 | **solid mass g:** 30.08 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** As exported; ensure on bed | **notes:** Spherical root eight rods muzzle rings integrated; normal auto supports; supplied servo horn; decorative arm only

9. part: 10_head_motor_carriage | **quantity:** 1 | **material:** PETG | **x mm:** 51.0 | **y mm:** 80.0 | **z mm:** 21.0 | **volume mm3:** 21985.9 | **solid mass g:** 27.92 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **brim mm per side:** 5 | **print rotation z degrees:** 0 | **h2d single nozzle with brim:** True | **pass check:** True | **orientation:** Floor down | **notes:** Radial slots padded TT saddle and cable-tie slots;4walls30%infill

Point-to-point wiring schedule

electronics\wiring.csv

1. **net:** BAT_POS | **origin:** B1.PP30+ | **destination:** F1.IN | **wire awg:** 18 | **color:** red | **note:** 7.5A fuse within 100mm of pack
2. **net:** BAT_FUSED | **origin:** F1.OUT | **destination:** S1.1 | **wire awg:** 18 | **color:** red
3. **net:** PACK_SW | **origin:** S1.3 | **destination:** S2.1 | **wire awg:** 18 | **color:** red | **note:** NKK S1A ON connects terminals1 and3
4. **net:** PACK_SW | **origin:** S1.3 | **destination:** F4.IN | **wire awg:** 18 | **color:** red | **note:** NKK S1A ON connects terminals1 and3
5. **net:** PACK_SW | **origin:** S1.3 | **destination:** R10.1 | **wire awg:** 18 | **color:** red | **note:** NKK S1A ON connects terminals1 and3
6. **net:** ACT_IN | **origin:** S2.3 | **destination:** F2.IN | **wire awg:** 18 | **color:** orange | **note:** Physical actuator power disconnect; logic stays powered
7. **net:** ACT_IN | **origin:** S2.3 | **destination:** F3.IN | **wire awg:** 18 | **color:** orange | **note:** Physical actuator power disconnect; logic stays powered
8. **net:** MOTOR_IN | **origin:** F2.OUT | **destination:** U2.VIN | **wire awg:** 18 | **color:** orange | **note:** F2 3A
9. **net:** SERVO_IN | **origin:** F3.OUT | **destination:** U3.VIN | **wire awg:** 18 | **color:** orange | **note:** F3 3A
10. **net:** LOGIC_IN | **origin:** F4.OUT | **destination:** U4.IN | **wire awg:** 22 | **color:** red | **note:** F4 1A
11. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** U5.VMotor | **wire awg:** 18 | **color:** orange
12. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** U6.VMotor | **wire awg:** 18 | **color:** orange
13. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** U11.VMotor | **wire awg:** 18 | **color:** orange
14. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** U9.14 | **wire awg:** 18 | **color:** orange
15. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** R12.1 | **wire awg:** 18 | **color:** orange
16. **net:** 5V_SERVO | **origin:** U3.VOUT | **destination:** J_SERVO.2 | **wire awg:** 18 | **color:** red | **note:** Four separate arm-servo branches; never feed through PCA9685 V+ traces
17. **net:** 5V_SERVO | **origin:** U3.VOUT | **destination:** U8.14 | **wire awg:** 18 | **color:** red | **note:** Four separate arm-servo branches; never feed through PCA9685 V+ traces
18. **net:** 5V_LOGIC | **origin:** U4.OUT | **destination:** J_USB_ISO.1 | **wire awg:** 22 | **color:** red
19. **net:** 5V_LOGIC | **origin:** U4.OUT | **destination:** U7.VIN | **wire awg:** 22 | **color:** red
20. **net:** 5V_LOGIC | **origin:** U4.OUT | **destination:** R14.1 | **wire awg:** 22 | **color:** red
21. **net:** ESP_5V | **origin:** J_USB_ISO.2 | **destination:** U1.5V | **wire awg:** 22 | **color:** red | **note:** Removable run-power link; OPEN and unplug battery before USB
22. **net:** 3V3 | **origin:** U1.3V3 | **destination:** U10.VCC | **wire awg:** 26 | **color:** red
23. **net:** 3V3 | **origin:** U1.3V3 | **destination:** R6.1 | **wire awg:** 26 | **color:** red
24. **net:** GND | **origin:** B1.PP30- | **destination:** U2.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board
25. **net:** GND | **origin:** B1.PP30- | **destination:** U3.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board
26. **net:** GND | **origin:** B1.PP30- | **destination:** U4.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board
27. **net:** GND | **origin:** B1.PP30- | **destination:** U5.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board
28. **net:** GND | **origin:** B1.PP30- | **destination:** U6.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board

29. **net:** GND | **origin:** B1.PP30- | **destination:** U11.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board

30. **net:** GND | **origin:** B1.PP30- | **destination:** J_SERVO.1 | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board

31. **net:** GND | **origin:** GROUND_STAR | **destination:** U1.GND | **wire awg:** 22 | **color:** black

32. **net:** GND | **origin:** GROUND_STAR | **destination:** U7.GND | **wire awg:** 22 | **color:** black

33. **net:** GND | **origin:** GROUND_STAR | **destination:** U8.7 | **wire awg:** 22 | **color:** black

34. **net:** GND | **origin:** GROUND_STAR | **destination:** U9.7 | **wire awg:** 22 | **color:** black

35. **net:** GND | **origin:** GROUND_STAR | **destination:** U10.GND | **wire awg:** 22 | **color:** black

36. **net:** GND | **origin:** B1.PP30- | **destination:** GROUND_STAR | **wire awg:** 18 | **color:** black

37. **net:** LEFT_IN1 | **origin:** U1.GPIO25 | **destination:** U5.AIN1 | **wire awg:** 26 | **color:** signal

38. **net:** LEFT_IN1 | **origin:** U1.GPIO25 | **destination:** U5.BIN1 | **wire awg:** 26 | **color:** signal

39. **net:** LEFT_IN2 | **origin:** U1.GPIO26 | **destination:** U5.AIN2 | **wire awg:** 26 | **color:** signal

40. **net:** LEFT_IN2 | **origin:** U1.GPIO26 | **destination:** U5.BIN2 | **wire awg:** 26 | **color:** signal

41. **net:** RIGHT_IN1 | **origin:** U1.GPIO32 | **destination:** U6.AIN1 | **wire awg:** 26 | **color:** signal

42. **net:** RIGHT_IN1 | **origin:** U1.GPIO32 | **destination:** U6.BIN1 | **wire awg:** 26 | **color:** signal

43. **net:** RIGHT_IN2 | **origin:** U1.GPIO33 | **destination:** U6.AIN2 | **wire awg:** 26 | **color:** signal

44. **net:** RIGHT_IN2 | **origin:** U1.GPIO33 | **destination:** U6.BIN2 | **wire awg:** 26 | **color:** signal

45. **net:** LEFT_IN1 | **origin:** LEFT_IN1 | **destination:** R1.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

46. **net:** GND | **origin:** GND | **destination:** R1.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

47. **net:** LEFT_IN2 | **origin:** LEFT_IN2 | **destination:** R2.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

48. **net:** GND | **origin:** GND | **destination:** R2.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

49. **net:** RIGHT_IN1 | **origin:** RIGHT_IN1 | **destination:** R3.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

50. **net:** GND | **origin:** GND | **destination:** R3.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

51. **net:** RIGHT_IN2 | **origin:** RIGHT_IN2 | **destination:** R4.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

52. **net:** GND | **origin:** GND | **destination:** R4.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

53. **net:** MOTOR_ENABLE | **origin:** MOTOR_ENABLE | **destination:** R5.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

54. **net:** GND | **origin:** GND | **destination:** R5.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

55. **net:** MOTOR_ENABLE | **origin:** U1.GPIO12 | **destination:** U5.SLP | **wire awg:** 26 | **color:** signal

56. **net:** MOTOR_ENABLE | **origin:** U1.GPIO12 | **destination:** U6.SLP | **wire awg:** 26 | **color:** signal

57. **net:** MOTOR_ENABLE | **origin:** U1.GPIO12 | **destination:** U11.SLP | **wire awg:** 26 | **color:** signal

58. **net:** SDA | **origin:** U1.GPIO21 | **destination:** U10.SDA | **wire awg:** 26 | **color:** signal

59. **net:** SCL | **origin:** U1.GPIO22 | **destination:** U10.SCL | **wire awg:** 26 | **color:** signal

60. **net:** SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U10.OE | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

61. **net:** SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** R6.2 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

62. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.1 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

63. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.4 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

64. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.10 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

65. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.13 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down; U9 uses separate direction enables

66. net: I2S_BCLK | **origin:** U1.GPIO17 | **destination:** U7.BCLK | **wire awg:** 26 | **color:** signal

67. net: I2S_LRC | **origin:** U1.GPIO13 | **destination:** U7.LRC | **wire awg:** 26 | **color:** signal

68. net: I2S_DATA | **origin:** U1.GPIO15 | **destination:** U7.DIN | **wire awg:** 26 | **color:** signal | **note:** GPIO15 strap: amplifier INPUT only; no pull-up/down added

69. net: BAT_ADC | **origin:** R10.2 | **destination:** R11.1 | **wire awg:** 26 | **color:** signal

70. net: BAT_ADC | **origin:** R10.2 | **destination:** U1.GPIO39 | **wire awg:** 26 | **color:** signal

71. net: BAT_ADC | **origin:** R10.2 | **destination:** C10.1 | **wire awg:** 26 | **color:** signal

72. net: GND | **origin:** GROUND_STAR | **destination:** R11.2 | **wire awg:** 26 | **color:** signal

73. net: GND | **origin:** GROUND_STAR | **destination:** C10.2 | **wire awg:** 26 | **color:** signal

74. net: ACT_OK | **origin:** R12.2 | **destination:** R13.1 | **wire awg:** 26 | **color:** signal

75. net: ACT_OK | **origin:** R12.2 | **destination:** U1.GPIO36 | **wire awg:** 26 | **color:** signal

76. net: ACT_OK | **origin:** R12.2 | **destination:** C11.1 | **wire awg:** 26 | **color:** signal

77. net: GND | **origin:** GROUND_STAR | **destination:** R13.2 | **wire awg:** 26 | **color:** signal

78. net: GND | **origin:** GROUND_STAR | **destination:** C11.2 | **wire awg:** 26 | **color:** signal

79. net: AMP_GAIN | **origin:** R14.2 | **destination:** U7.GAIN | **wire awg:** 26 | **color:** signal | **note:** R14 100k to5V_LOGIC sets3dB

80. net: SPK_P | **origin:** U7.OUT+ | **destination:** SPK1.+ | **wire awg:** 24 | **color:** yellow | **note:** Floating bridge output; NEVER ground speaker leads

81. net: SPK_N | **origin:** U7.OUT- | **destination:** SPK1.- | **wire awg:** 24 | **color:** blue

82. net: LEFT_FRONT_+ | **origin:** U5.AOUT1 | **destination:** M_LEFT_FRONT.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

83. net: LEFT_FRONT_- | **origin:** U5.AOUT2 | **destination:** M_LEFT_FRONT.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

84. net: LEFT_FRONT_+ | **origin:** LEFT_FRONT_+ | **destination:** C_LEFT_FRONT.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

85. net: LEFT_FRONT_- | **origin:** LEFT_FRONT_- | **destination:** C_LEFT_FRONT.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

86. net: LEFT_REAR_+ | **origin:** U5.BOUT1 | **destination:** M_LEFT_REAR.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

87. net: LEFT_REAR_- | **origin:** U5.BOUT2 | **destination:** M_LEFT_REAR.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

88. net: LEFT_REAR_+ | **origin:** LEFT_REAR_+ | **destination:** C_LEFT_REAR.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

89. net: LEFT_REAR_- | **origin:** LEFT_REAR_- | **destination:** C_LEFT_REAR.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

90. net: RIGHT_FRONT_+ | **origin:** U6.AOUT1 | **destination:** M_RIGHT_FRONT.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

91. net: RIGHT_FRONT_- | **origin:** U6.AOUT2 | **destination:** M_RIGHT_FRONT.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

92. net: RIGHT_FRONT_+ | **origin:** RIGHT_FRONT_+ | **destination:** C_RIGHT_FRONT.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

93. net: RIGHT_FRONT_- | **origin:** RIGHT_FRONT_- | **destination:** C_RIGHT_FRONT.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

94. net: RIGHT_REAR_+ | **origin:** U6.BOUT1 | **destination:** M_RIGHT_REAR.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

95. net: RIGHT_REAR_- | **origin:** U6.BOUT2 | **destination:** M_RIGHT_REAR.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

96. net: RIGHT_REAR_+ | **origin:** RIGHT_REAR_+ | **destination:** C_RIGHT_REAR.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

97. net: RIGHT_REAR_- | **origin:** RIGHT_REAR_- | **destination:** C_RIGHT_REAR.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

98. net: PWM0 | **origin:** U10.PWM0 | **destination:** U8.2 | **wire awg:** 26 | **color:** signal

99. net: PWM0 | **origin:** PWM0 | **destination:** R20.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

100. net: GND | **origin:** GND | **destination:** R20.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

101. net: PWM5V_0 | **origin:** U8.3 | **destination:** R30.1 | **wire awg:** 26 | **color:** signal

102. net: SERVO_LEFT_YAW | **origin:** R30.2 | **destination:** SV0.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

103. net: SERVO_LEFT_YAW | **origin:** SERVO_LEFT_YAW | **destination:** R40.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

104. net: GND | **origin:** GND | **destination:** R40.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

105. net: 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV0.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

106. net: GND | **origin:** J_SERVO.1 | **destination:** SV0.GND | **wire awg:** 22 | **color:** black/brown

107. net: PWM1 | **origin:** U10.PWM1 | **destination:** U8.5 | **wire awg:** 26 | **color:** signal

108. net: PWM1 | **origin:** PWM1 | **destination:** R21.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

109. net: GND | **origin:** GND | **destination:** R21.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

110. net: PWM5V_1 | **origin:** U8.6 | **destination:** R31.1 | **wire awg:** 26 | **color:** signal

111. net: SERVO_LEFT_PITCH | **origin:** R31.2 | **destination:** SV1.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

112. net: SERVO_LEFT_PITCH | **origin:** SERVO_LEFT_PITCH | **destination:** R41.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

113. net: GND | **origin:** GND | **destination:** R41.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

114. net: 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV1.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

115. net: GND | **origin:** J_SERVO.1 | **destination:** SV1.GND | **wire awg:** 22 | **color:** black/brown

116. net: PWM2 | **origin:** U10.PWM2 | **destination:** U8.9 | **wire awg:** 26 | **color:** signal

117. net: PWM2 | **origin:** PWM2 | **destination:** R22.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

118. net: GND | **origin:** GND | **destination:** R22.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

119. net: PWM5V_2 | **origin:** U8.8 | **destination:** R32.1 | **wire awg:** 26 | **color:** signal

120. net: SERVO_RIGHT_YAW | **origin:** R32.2 | **destination:** SV2.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

121. net: SERVO_RIGHT_YAW | **origin:** SERVO_RIGHT_YAW | **destination:** R42.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

122. net: GND | **origin:** GND | **destination:** R42.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

123. net: 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV2.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

124. net: GND | **origin:** J_SERVO.1 | **destination:** SV2.GND | **wire awg:** 22 | **color:** black/brown

125. net: PWM3 | **origin:** U10.PWM3 | **destination:** U8.12 | **wire awg:** 26 | **color:** signal

126. net: PWM3 | **origin:** PWM3 | **destination:** R23.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

127. net: GND | **origin:** GND | **destination:** R23.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

128. net: PWM5V_3 | **origin:** U8.11 | **destination:** R33.1 | **wire awg:** 26 | **color:** signal

129. net: SERVO_RIGHT_PITCH | **origin:** R33.2 | **destination:** SV3.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

130. net: SERVO_RIGHT_PITCH | **origin:** SERVO_RIGHT_PITCH | **destination:** R43.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

131. net: GND | **origin:** GND | **destination:** R43.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

132. net: 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV3.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

133. net: GND | **origin:** J_SERVO.1 | **destination:** SV3.GND | **wire awg:** 22 | **color:** black/brown

134. net: HEAD_PWM | **origin:** U1.GPIO2 | **destination:** U9.2 | **wire awg:** 26 | **color:** signal | **note:** 20kHz 8bit PWM; low before direction updates; cap100/255

135. net: HEAD_PWM | **origin:** U1.GPIO2 | **destination:** U9.5 | **wire awg:** 26 | **color:** signal | **note:** 20kHz 8bit PWM; low before direction updates; cap100/255

136. net: HEAD_PWM | **origin:** HEAD_PWM | **destination:** R24.1 | **wire awg:** component | **note:** 10k pull-down: GPIO2 boot-safe and head PWM off on reset

137. net: GND | **origin:** GND | **destination:** R24.2 | **wire awg:** component | **note:** 10k pull-down: GPIO2 boot-safe and head PWM off on reset

138. net: HEAD_DISABLE1 | **origin:** U10.PWM4 | **destination:** U9.1 | **wire awg:** 26 | **color:** signal | **note:** STATIC HIGH disables this gate; LOW enables; never servo pulses

139. net: HEAD_DISABLE1 | **origin:** HEAD_DISABLE1 | **destination:** R7.1 | **wire awg:** component | **note:** 10k pull-up to3V3; AHCT TTL threshold

140. net: 3V3 | **origin:** 3V3 | **destination:** R7.2 | **wire awg:** component | **note:** 10k pull-up to3V3; AHCT TTL threshold

141. net: HEAD_BUFFER1 | **origin:** U9.3 | **destination:** R34.1 | **wire awg:** 26 | **color:** signal

142. net: HEAD_IN1 | **origin:** R34.2 | **destination:** U11.AIN1 | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor; only one input gets PWM

143. net: HEAD_IN1 | **origin:** HEAD_IN1 | **destination:** R44.1 | **wire awg:** component | **note:** 10k pull-down at driver input

144. net: GND | **origin:** GND | **destination:** R44.2 | **wire awg:** component | **note:** 10k pull-down at driver input

145. net: HEAD_DISABLE2 | **origin:** U10.PWM5 | **destination:** U9.4 | **wire awg:** 26 | **color:** signal | **note:** STATIC HIGH disables this gate; LOW enables; never servo pulses

146. net: HEAD_DISABLE2 | **origin:** HEAD_DISABLE2 | **destination:** R8.1 | **wire awg:** component | **note:** 10k pull-up to3V3; AHCT TTL threshold

147. net: 3V3 | **origin:** 3V3 | **destination:** R8.2 | **wire awg:** component | **note:** 10k pull-up to3V3; AHCT TTL threshold

148. net: HEAD_BUFFER2 | **origin:** U9.6 | **destination:** R35.1 | **wire awg:** 26 | **color:** signal

149. net: HEAD_IN2 | **origin:** R35.2 | **destination:** U11.AIN2 | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor; only one input gets PWM

150. net: HEAD_IN2 | **origin:** HEAD_IN2 | **destination:** R45.1 | **wire awg:** component | **note:** 10k pull-down at driver input

151. net: GND | **origin:** GND | **destination:** R45.2 | **wire awg:** component | **note:** 10k pull-down at driver input

152. net: GND | **origin:** GROUND_STAR | **destination:** U11.BIN1 | **wire awg:** 26 | **color:** signal | **note:** Unused bridge B inputs low; BOUT1/BOUT2 unconnected

153. net: GND | **origin:** GROUND_STAR | **destination:** U11.BIN2 | **wire awg:** 26 | **color:** signal | **note:** Unused bridge B inputs low; BOUT1/BOUT2 unconnected

154. net: HEAD_MOTOR+ | **origin:** U11.AOUT1 | **destination:** M_HEAD.+ | **wire awg:** 22 | **color:** yellow

155. net: HEAD_MOTOR- | **origin:** U11.AOUT2 | **destination:** M_HEAD.- | **wire awg:** 22 | **color:** blue

156. net: HEAD_MOTOR+ | **origin:** HEAD_MOTOR+ | **destination:** C_HEAD.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

157. net: HEAD_MOTOR- | **origin:** HEAD_MOTOR- | **destination:** C_HEAD.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

158. net: GND | **origin:** GROUND_STAR | **destination:** U9.9 | **wire awg:** 26 | **color:** black | **note:** Unused buffer INPUT tied low

159. net: GND | **origin:** GROUND_STAR | **destination:** U9.12 | **wire awg:** 26 | **color:** black | **note:** Unused buffer INPUT tied low

160. net: 5V_MOTOR | **origin:** U2.VOUT | **destination:** U9.10 | **wire awg:** 26 | **color:** orange | **note:** Unused buffer OE high

161. net: 5V_MOTOR | **origin:** U2.VOUT | **destination:** U9.13 | **wire awg:** 26 | **color:** orange | **note:** Unused buffer OE high

162. net: MOTOR_IN | **origin:** MOTOR_IN | **destination:** C1.1 | **wire awg:** component | **note:** 220uF 25V low-ESR

163. net: GND | **origin:** GND | **destination:** C1.2 | **wire awg:** component | **note:** 220uF 25V low-ESR

164. net: SERVO_IN | **origin:** SERVO_IN | **destination:** C2.1 | **wire awg:** component | **note:** 220uF 25V low-ESR

165. net: GND | **origin:** GND | **destination:** C2.2 | **wire awg:** component | **note:** 220uF 25V low-ESR

166. net: LOGIC_IN | **origin:** LOGIC_IN | **destination:** C3.1 | **wire awg:** component | **note:** 100uF 25V

167. net: GND | **origin:** GND | **destination:** C3.2 | **wire awg:** component | **note:** 100uF 25V

168. net: 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** C4.1 | **wire awg:** component | **note:** 470uF 10V low-ESR

169. net: GND | **origin:** GND | **destination:** C4.2 | **wire awg:** component | **note:** 470uF 10V low-ESR

170. net: 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** C5.1 | **wire awg:** component | **note:** 470uF 10V low-ESR

171. net: GND | **origin:** GND | **destination:** C5.2 | **wire awg:** component | **note:** 470uF 10V low-ESR

172. net: 5V_SERVO | **origin:** 5V_SERVO | **destination:** C6.1 | **wire awg:** component | **note:** 1000uF 10V low-ESR

173. net: GND | **origin:** GND | **destination:** C6.2 | **wire awg:** component | **note:** 1000uF 10V low-ESR

174. net: 5V_LOGIC | **origin:** 5V_LOGIC | **destination:** C7.1 | **wire awg:** component | **note:** 470uF 10V low-ESR

175. net: GND | **origin:** GND | **destination:** C7.2 | **wire awg:** component | **note:** 470uF 10V low-ESR

176. net: 5V_SERVO | **origin:** 5V_SERVO | **destination:** C8.1 | **wire awg:** component | **note:** 100nF ceramic

177. net: GND | **origin:** GND | **destination:** C8.2 | **wire awg:** component | **note:** 100nF ceramic

178. net: 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** C9.1 | **wire awg:** component | **note:** 100nF ceramic

179. **net:** GND | **origin:** GND | **destination:** C9.2 | **wire awg:** component | **note:** 100nF ceramic

180. **net:** 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** C12.1 | **wire awg:** component | **note:** 470uF 10V low-ESR at head driver

181. **net:** GND | **origin:** GND | **destination:** C12.2 | **wire awg:** component | **note:** 470uF 10V low-ESR at head driver

182. **net:** 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** R15.1 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

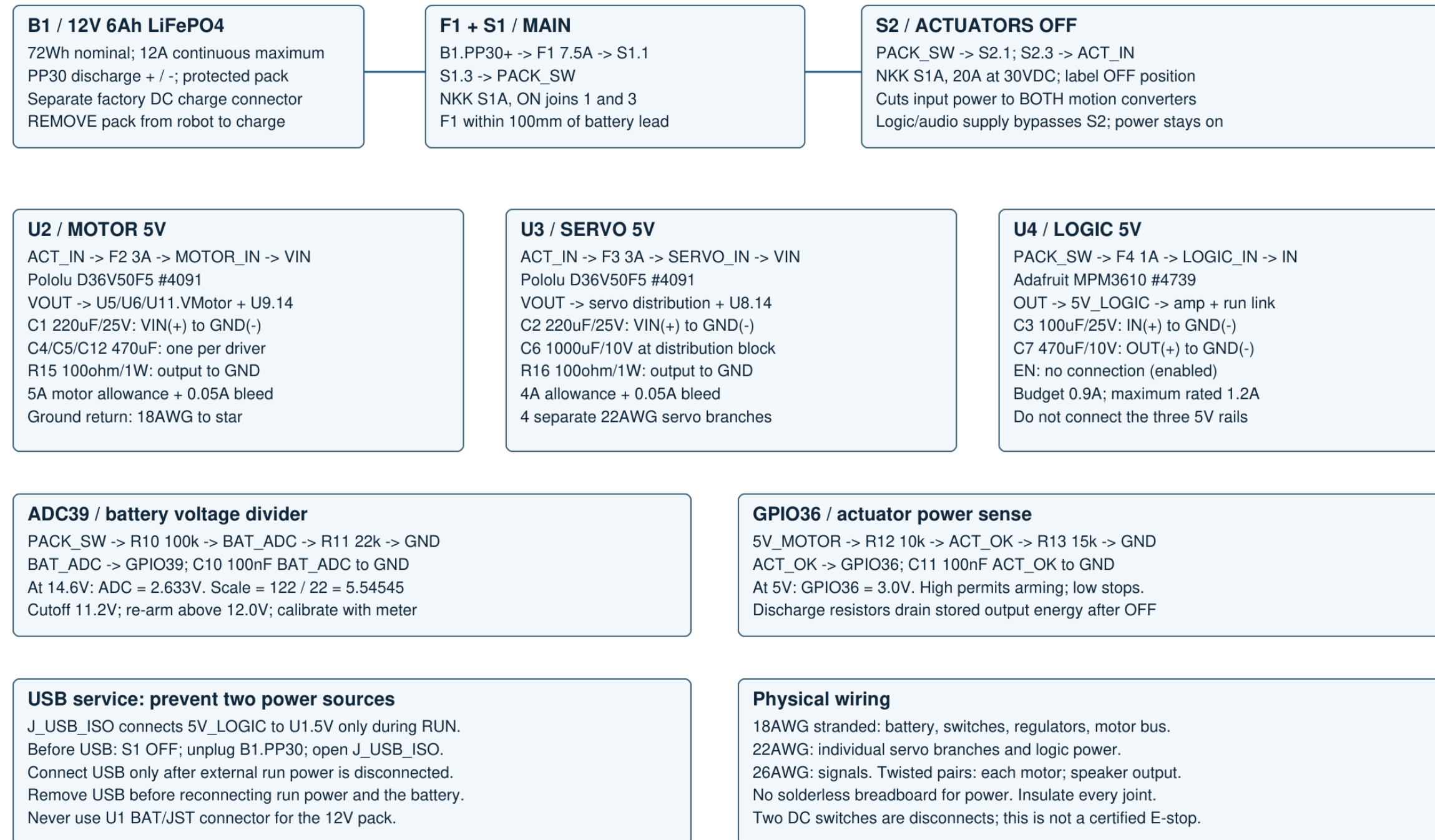
183. **net:** GND | **origin:** GND | **destination:** R15.2 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

184. **net:** 5V_SERVO | **origin:** 5V_SERVO | **destination:** R16.1 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

185. **net:** GND | **origin:** GND | **destination:** R16.2 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

01 / Power distribution and isolation

B1 = Bioenno BLF-1206A + matched BPC-1502DC charger. All grounds join at GROUND_STAR.



02 / Controller, drive motors and sound

U1 = original LILYGO TTGO T-Display ESP32, 1.14 inch ST7789. GPIO numbers, not header positions.

U1 / exact external pin assignment

GPIO21	_____	SDA -> U10.SDA
GPIO22	_____	SCL -> U10.SCL
GPIO27	_____	SERVO_DISABLE / OE
GPIO12	_____	MOTOR_ENABLE / SLP
GPIO25	_____	LEFT_IN1
GPIO26	_____	LEFT_IN2
GPIO32	_____	RIGHT_IN1
GPIO33	_____	RIGHT_IN2
GPIO17	_____	I2S_BCLK
GPIO13	_____	I2S_LRC
GPIO15	_____	I2S_DATA
GPIO39 (SVN)	_____	BAT_ADC
GPIO36 (SVP)	_____	ACT_OK
GPIO2	_____	HEAD_PWM -> U9.2 + .5

Reserved board connections

TFT: MOSI19, SCLK18, CS5, DC16, RST23, BL4
Buttons: GPIO0 and GPIO35; reserve both.
Onboard battery ADC34 and power14 are NOT pack sense.
GPIO12 and GPIO2 each have a 10k boot-safe pull-down.
GPIO15 connects only to amplifier DIN; no external bias.
GPIO37, 38 and UART pins: no external connection.

Hardware stopped on reset

R1..R4: 10k from each wheel input net to GND.
R5: 10k MOTOR_ENABLE to GND; ALL THREE SLP pins join.
R6: 10k SERVO_DISABLE to3V3; R24: HEAD_PWM to GND.
Firmware: SLP low, PWM low, OE high before devices init.
U1.3V3 -> U10.VCC; all GND pins join common ground.
Physical power OFF removes wheel, head and arm power.

U5 / Adafruit DRV8833 #3297

LEFT_IN1 -> AIN1 AND BIN1 (INPUTS ONLY)
LEFT_IN2 -> AIN2 AND BIN2 (INPUTS ONLY)
SLP <- MOTOR_ENABLE; VMotor <- 5V_MOTOR; GND <- star
AOUT1/AOUT2 -> M_LEFT_FRONT +/-
BOUT1/BOUT2 -> M_LEFT_REAR +/-
Keep BOTH default0.2ohm sense resistors:1A per bridge.
DO NOT parallel the output bridges or two motors on one.
FLT, VM, ASEN, BSEN: no external connection.

U6 / Adafruit DRV8833 #3297

RIGHT_IN1 -> AIN1 AND BIN1 (INPUTS ONLY)
RIGHT_IN2 -> AIN2 AND BIN2 (INPUTS ONLY)
SLP <- MOTOR_ENABLE; VMotor <- 5V_MOTOR; GND <- star
AOUT1/AOUT2 -> M_RIGHT_FRONT +/-
BOUT1/BOUT2 -> M_RIGHT_REAR +/-
Keep BOTH default0.2ohm sense resistors:1A per bridge.
DO NOT parallel the output bridges or two motors on one.
FLT, VM, ASEN, BSEN: no external connection.

U7 / MAX98357A + SPK1

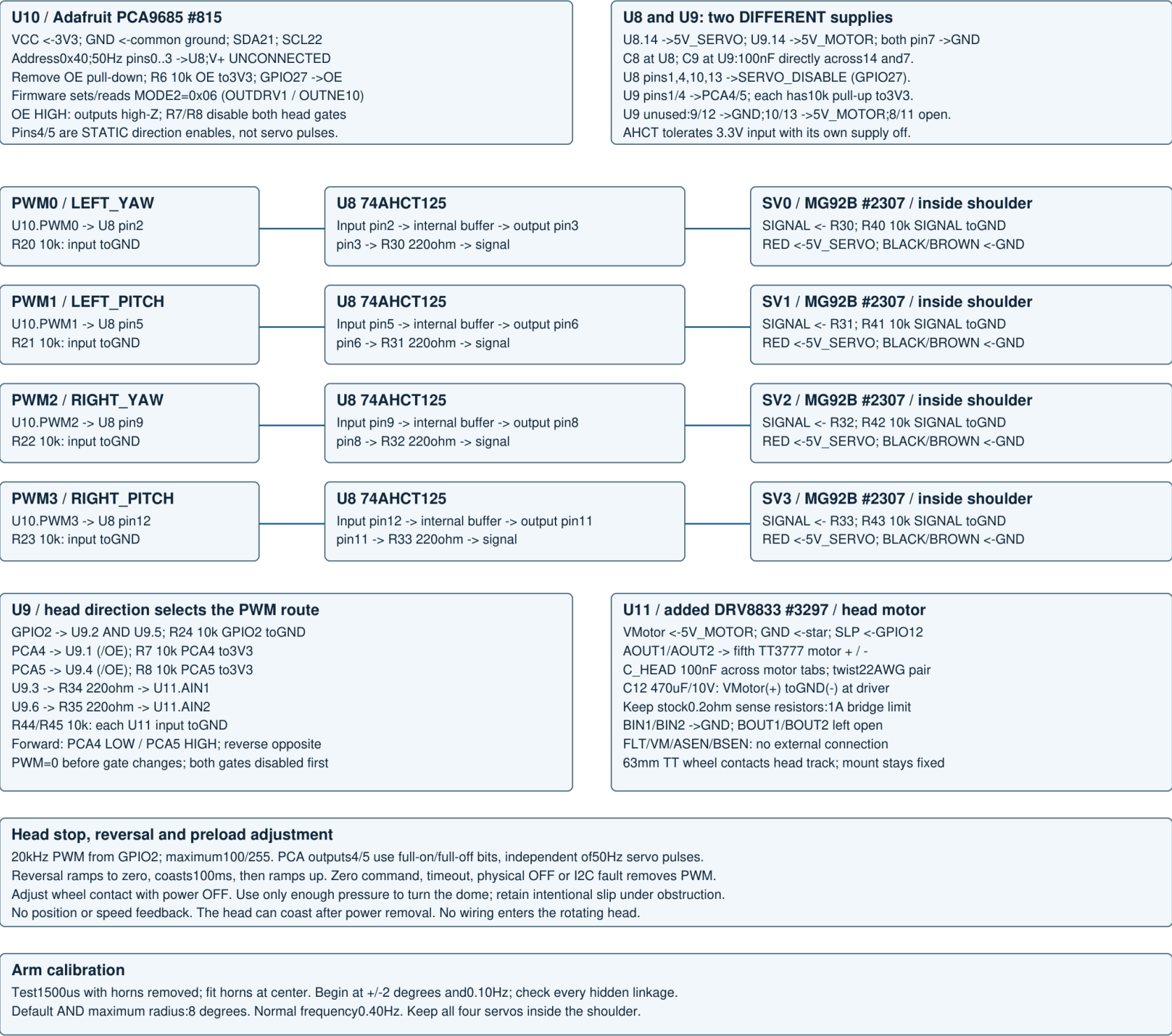
VIN <- 5V_LOGIC; GND <- common ground
BCLK <- GPIO17; LRC <- GPIO13; DIN <- GPIO15
GAIN -> R14 100k -> 5V_LOGIC (3dB)
SD: leave factory 1Mohm resistor configuration (mono mix)
OUT+ -> SPK1+; OUT- -> SPK1-; NEITHER lead to GND
SPK1: Adafruit 1313, 8ohm 1W; fixed digital gain 0.28
No external DAC, SD card or MP3 trigger module required.

Drive commissioning

100nF ceramic across each motor's metal tabs.
Use fast-decay PWM: one input PWM, other input low.
Stop both inputs low and SLP low; this is coast, not a brake.
Check each motor forward direction with wheels in air.
Swap that motor's two leads if its direction is wrong.
1A limit protects the driver, not indefinite motor stalls.
Indoor flat surface; no carpet, stairs or sustained wheel jams.

03 / Four arm servos and head PWM routing

All chip numbers below are DIP-14 PHYSICAL pin numbers. U8/U9 are 74AHCT125, not 74HC125.



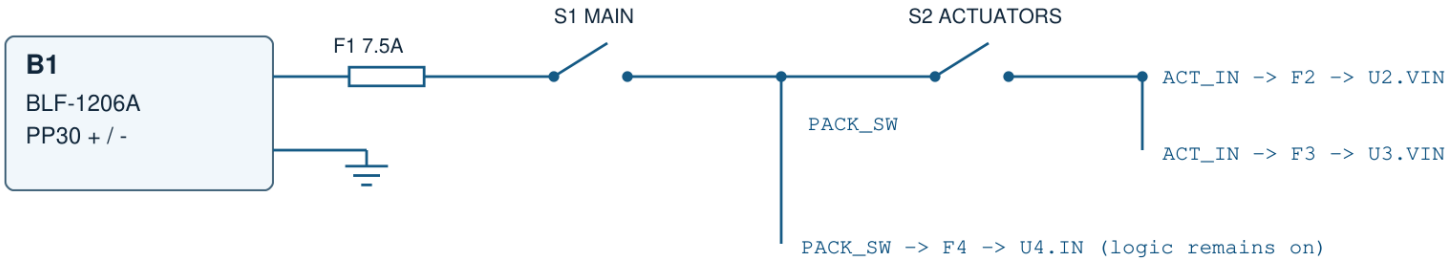
DALEK ROUND / FRICTION | 2026-09-12 | Circuit + wiring.csv form one drawing set | Physical validation required

electronics\03-servos.svg | Drawing page: 17 x 11 inches. Use named nets and the wiring schedule together.

04 / Circuit details: power, sensing and bridge outputs

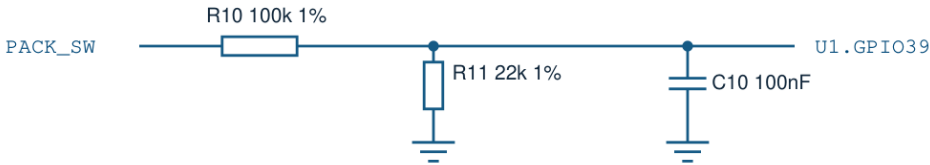
Conventional connection symbols. Dots are junctions. Matching net labels join all four sheets.

A. Battery fuse and independent motion disconnect



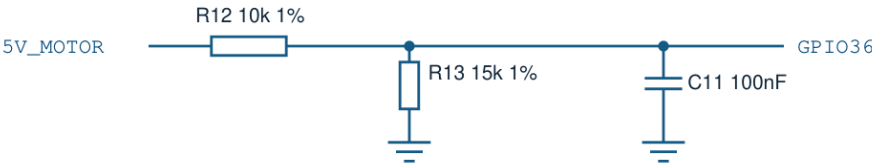
S1/S2 are shown OFF. NKK S1A ON joins terminals1 and3. Ground is never switched.

B. Battery measurement / GPIO39



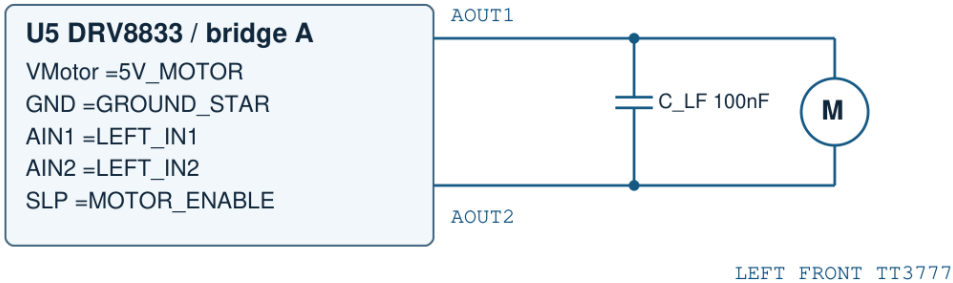
$14.6V \times 22 / (100 + 22) = 2.633V$; firmware multiplier5.54545

C. Motion power sense / GPIO36



$5V \times 15 / (10 + 15) = 3.0V$; low means physical power OFF

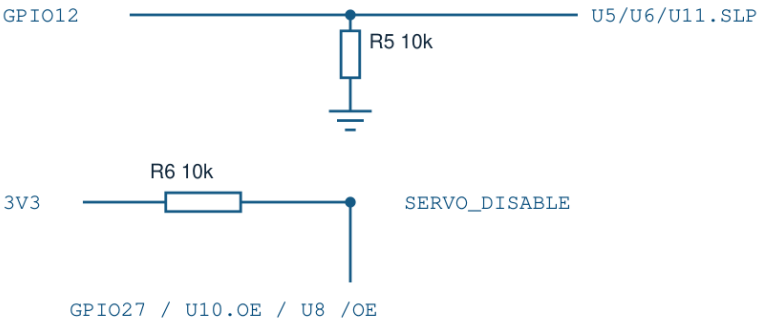
D. Motor output circuit (repeat for all5 motors)



U5.B -> LEFT REAR; U6.A/B -> RIGHT FRONT/REAR; U11.A -> HEAD (routing on sheet03).

Bridge outputs never join one another. The only shared connections are INPUTS and supplies.

E. Reset default states



Remove originalPCA0E pull-down.

GPIO2 and head AIN1/AIN2:10k pull-downs.

POWER NOTES: F2/F3=3A; F4=1A. Correct capacitor polarity. All connection values also appear in sheets01-03.

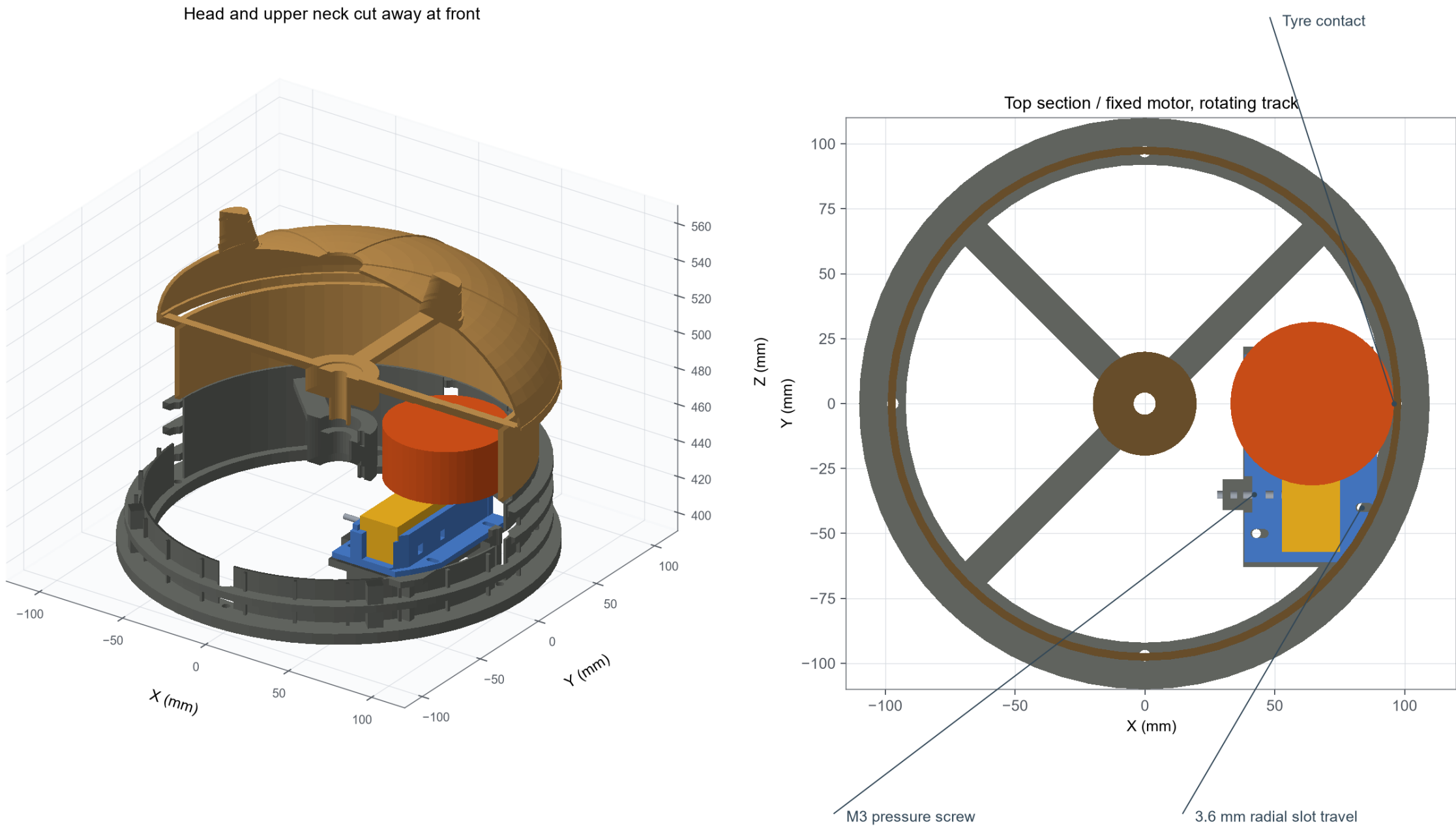
DALEK ROUND / FRICTION | 2026-09-12 | Circuit + wiring.csv form one drawing set | Physical validation required

electronics\04-circuit-details.svg | Drawing page: 17 x 11 inches. Use named nets and the wiring schedule together.

Adjustable head friction drive

Dalek ROUND-9 / adjustable head friction drive

Fifth TT3777 motor + 3766 wheel | Actual neck, carriage and head STL sections | Power off before adjusting



Blue: sliding carriage | Orange: 63 mm tyre | Yellow: motor envelope | Set light contact; preserve slip if obstructed.

Actual printed geometry shown in cutaway. Follow the mechanical adjustment and locking sequence with power isolated.